

MTH251 & MTH252

Mathematical Analysis I & II

Course Textbook

Department of Mathematical Sciences
UNIST

Mathematical Analysis I & II

Course Textbook

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Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

$$(P \vee Q) \wedge R$$

is evaluated by first computing $P \vee Q$.

- The **logical connectives** are interpreted in the order

$$\neg > \wedge > \vee > \implies > \iff$$

Hence

$$\neg P \wedge Q \vee R \implies S \iff T$$

is understood as

$$(((\neg P) \wedge Q) \vee R) \implies S) \iff T$$

- A **quantifier** binds the entire formula that follows it. Thus

$$\forall x, P(x) \wedge Q(x) \implies R(x)$$

means

$$\forall x ((P(x) \wedge Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

$$\begin{array}{c} P \\ \Downarrow \\ Q \end{array}$$

merely expresses that P is true, and therefore Q is true in the current discussion; it is not itself a proposition.

Chapter 1

The Number Systems

1.1 The Natural Number Systems

The set $\{1, 2, 3, \dots\}$ of **natural numbers** is denoted by $\mathbb{N}$, and is defined by the following axioms:

- $1 \in \mathbb{N}$
- $\forall n \in \mathbb{N}, s(n) \in \mathbb{N}$
- $\nexists n \in \mathbb{N}, s(n) = 1$
- $\forall m, n \in \mathbb{N}, s(m) = s(n) \implies m = n$
- If $S \subseteq \mathbb{N}$ satisfies
 - $1 \in S$
 - $\forall n \in S, s(n) \in S$

then $S = \mathbb{N}$

where $s(n)$ is called the **successor** of n . These axioms are called the **Peano axioms** (or *Peano Postulates*).

Remark. The **Peano axioms** do not guarantee the *existence* of $\mathbb{N}$, which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** is not discussed, so we will just assume that $\mathbb{N}$ exists.

Definition 1.1.1. Let $P(n)$ be a statement about $n \in \mathbb{N}$. Define

$$S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\} \subseteq \mathbb{N}$$

If $P(n)$ satisfies

- $P(1)$ is true
- $\forall n \in \mathbb{N}, P(n) \text{ is true} \implies P(s(n)) \text{ is true}$

Then S satisfies

- $1 \in S$
- $\forall n \in S, s(n) \in S$

Therefore,

$$\begin{array}{c} S = \mathbb{N} \\ \Downarrow \\ \forall n \in \mathbb{N}, P(n) \text{ is true} \end{array}$$

This method of proof is called the **principle of mathematical induction**.

Definition 1.1.2. Let $n, m \in \mathbb{N}$ be given.

The **addition** $+$ on $\mathbb{N}$ is defined as

- $n + 1 := s(n)$
- $n + s(m) := s(n + m)$

The **multiplication** $\cdot$ on $\mathbb{N}$ is defined as

- $n \cdot 1 := n$
- $n \cdot s(m) := n \cdot m + n$

Remark. Strictly speaking, [Definition 1.1.2](#) requires the **recursion theorem** to be strictly defined. In this textbook, the **recursion theorem** is not discussed, so we will just assume that the operations on $\mathbb{N}$ are **well-defined**.

Theorem 1.1.3. Let $n, m, k \in \mathbb{N}$ be given. Then the following properties hold:

- $n + m = m + n$
- $(n + m) + k = n + (m + k)$
- $n \cdot m = m \cdot n$
- $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- $n \cdot (m + k) = n \cdot m + n \cdot k$

Proof. We will not discuss the proof of this theorem, since it relies on the recursive definitions. $\square$

Theorem 1.1.4. $\mathbb{N}$ has the *well-ordering property*, which states that

$$\forall S \subseteq \mathbb{N}, S \neq \emptyset \implies \exists m \in S, m = \min S$$

Proof. Assume that

$$\exists S \subseteq \mathbb{N}, S \neq \emptyset \wedge (\nexists m \in S, m = \min S)$$

Define

- $T := \mathbb{N} \setminus S \subseteq \mathbb{N}$
- $L := \{n \in \mathbb{N} \mid \{1, 2, \dots, n\} \subseteq T\} \subseteq \mathbb{N}$

We will show that

$$\begin{aligned} L &= \mathbb{N} \\ \Downarrow \\ T &= \mathbb{N} \\ \Downarrow \\ S &= \emptyset \\ \Downarrow \\ &\perp \end{aligned}$$

- **Base case:** If $1 \in S$, then

$$\begin{aligned} 1 &\in S \\ \Downarrow \\ 1 &= \min S \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\begin{aligned} 1 &\notin S \\ \Downarrow \\ 1 &\in T \\ \Downarrow \\ 1 &\in L \end{aligned}$$

- **Inductive step:** Let $n \in L$ be given. Then

$$\begin{aligned} n &\in L \\ \Downarrow \\ \{1, 2, \dots, n\} &\subseteq T \\ \Downarrow \\ \{1, 2, \dots, n\} \cap S &= \emptyset \end{aligned}$$

If $n + 1 \in S$, then

$$\begin{aligned} n + 1 &\in S \\ \Downarrow \\ n + 1 &= \min S \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\begin{aligned} n + 1 &\notin S \\ \Downarrow \\ n + 1 &\in T \\ \Downarrow \\ \{1, 2, \dots, n + 1\} &\subseteq T \\ \Downarrow \\ n + 1 &\in L \end{aligned}$$

By the principle of mathematical induction, $L = \mathbb{N}$. □

1.2 Relations

Let S be a set. A **relation** $\sim$ on S is a subset of $S \times S$. Equivalently, a subset R of $S \times S$ is a relation on S . That is, for every $x, y \in S$,

- $R = \{(x, y) \in S \times S \mid x \sim y\}$
- $x \sim y \iff (x, y) \in R$

Definition 1.2.1. Let S be a set. A relation $\sim$ on S is called an **equivalence relation** if it satisfies the following properties:

- **Reflexivity:** $\forall x \in S, x \sim x$
- **Symmetry:** $\forall x, y \in S, x \sim y \implies y \sim x$
- **Transitivity:** $\forall x, y, z \in S, x \sim y \wedge y \sim z \implies x \sim z$

Definition 1.2.2. Let S be a set. A collection $\{A_\alpha\}_{\alpha \in I}$ of sets is called a **partition** of S if

- $\forall \alpha \in I, A_\alpha \neq \emptyset \wedge A_\alpha \subseteq S$
- $\forall \alpha, \beta \in I, \alpha \neq \beta \implies A_\alpha \cap A_\beta = \emptyset$
- $\bigcup_{\alpha \in I} A_\alpha = S$

Definition 1.2.3. Let S be a set, and let $\sim$ be an equivalence relation on S . For each $x \in S$, the **equivalence class** of x is defined as

$$[x] := \{y \in S \mid y \sim x\}$$

We denote the set of all equivalence classes of S by

$$S / \sim := \{[x] \mid x \in S\}$$

Theorem 1.2.4. Let S be a set, and let $\sim$ be an equivalence relation on S . Then

$$\forall x, y \in S, [x] \cap [y] \neq \emptyset \implies [x] = [y]$$

Proof. If $[x] \cap [y] \neq \emptyset$, then

$$\begin{aligned} \exists z \in S, z \in [x] \cap [y] \\ \Downarrow \\ z \in [x] \wedge z \in [y] \\ \Downarrow \\ z \sim x \wedge z \sim y \\ \Downarrow \\ x \sim z \wedge z \sim y \\ \Downarrow \\ x \sim y \end{aligned}$$

Therefore, for every $w \in [x]$,

$$\begin{aligned} w \sim x \wedge x \sim y \\ \Downarrow \\ w \sim y \\ \Downarrow \\ w \in [y] \end{aligned}$$

Therefore,

$$[x] \subseteq [y]$$

Similarly,

$$[y] \subseteq [x]$$

□

Theorem 1.2.5. Let S be a set, and let $\{A_\alpha\}_{\alpha \in I}$ be a partition of S . Define

$$x \sim y \iff \exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$$

for all $x, y \in S$. Then

(A) $\sim$ is an equivalence relation on S

(B) $\{A_\alpha\}_{\alpha \in I} = S / \sim$

Proof. (A) Let $x, y, z \in S$ be given.

• **Reflexivity:** Since $x \in S$,

$$\begin{aligned} x &\in S \\ &\Downarrow \\ x &\in \bigcup_{\alpha \in I} A_\alpha \\ &\Downarrow \\ \exists \alpha \in I, x &\in A_\alpha \\ &\Downarrow \\ x &\sim x \end{aligned}$$

• **Symmetry:** If $x \sim y$, then

$$\begin{aligned} \exists \alpha \in I, x &\in A_\alpha \wedge y \in A_\alpha \\ &\Downarrow \\ y &\in A_\alpha \wedge x \in A_\alpha \\ &\Downarrow \\ y &\sim x \end{aligned}$$

• **Transitivity:** If $x \sim y \wedge y \sim z$, then

- $\exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$
- $\exists \beta \in I, y \in A_\beta \wedge z \in A_\beta$

Therefore,

$$\begin{aligned} y &\in A_\alpha \wedge y \in A_\beta \\ &\Downarrow \\ A_\alpha \cap A_\beta &\neq \emptyset \\ &\Downarrow \\ \alpha &= \beta \\ &\Downarrow \\ x &\in A_\alpha \wedge z \in A_\alpha \\ &\Downarrow \\ x &\sim z \end{aligned}$$

(B) Let $x \in S$ be given. Then

$$\begin{aligned} x &\in S \\ &\Downarrow \\ x &\in \bigcup_{\alpha \in I} A_\alpha \\ &\Downarrow \\ \exists \alpha \in I, x &\in A_\alpha \end{aligned}$$

We will show that $A_\alpha = [x]$.

- ($\subseteq$) Let $y \in A_\alpha$ be given. Then

$$\begin{aligned} y \in A_\alpha \wedge x \in A_\alpha \\ \Downarrow \\ y \sim x \\ \Downarrow \\ y \in [x] \end{aligned}$$

- ($\supseteq$) Let $y \in [x]$ be given. Then

$$\begin{aligned} y \sim x \\ \Downarrow \\ \exists \beta \in I, y \in A_\beta \wedge x \in A_\beta \\ \Downarrow \\ x \in A_\alpha \wedge x \in A_\beta \\ \Downarrow \\ x \in A_\alpha \cap A_\beta \\ \Downarrow \\ A_\alpha \cap A_\beta \neq \emptyset \\ \Downarrow \\ \alpha = \beta \\ \Downarrow \\ y \in A_\alpha \end{aligned}$$

To sum up,

$$\begin{aligned} \forall x \in S, \exists \alpha \in I, A_\alpha = [x] \\ \Downarrow \\ S / \sim \subseteq \{A_\alpha\}_{\alpha \in I} \end{aligned}$$

Also,

$$\begin{aligned} \forall \alpha \in I, A_\alpha \neq \emptyset \\ \Downarrow \\ \exists x \in A_\alpha \subseteq S \\ \Downarrow \\ A_\alpha = [x] \\ \Downarrow \\ \{A_\alpha\}_{\alpha \in I} \subseteq S / \sim \end{aligned}$$

Therefore,

$$\{A_\alpha\}_{\alpha \in I} = S / \sim$$

□

1.3 The Integer and Rational Number Systems

Define a relation $\sim$ on $\mathbb{N} \times \mathbb{N}$ as

$$(a, b) \sim (c, d) \iff a + d = b + c$$

Let $(a, b), (c, d), (e, f) \in \mathbb{N} \times \mathbb{N}$ be given. By [Theorem 1.1.3](#),

- **Reflexivity:** Since $a \in \mathbb{N} \wedge b \in \mathbb{N}$,

$$\begin{aligned} a + b &= b + a \\ \Downarrow \\ (a, b) &\sim (a, b) \end{aligned}$$

- **Symmetry:** If $(a, b) \sim (c, d)$, then

$$\begin{aligned} a + d &= b + c \\ \Downarrow \\ c + b &= d + a \\ \Downarrow \\ (c, d) &\sim (a, b) \end{aligned}$$

- **Transitivity:** If $(a, b) \sim (c, d) \wedge (c, d) \sim (e, f)$, then

$$\begin{aligned} a + d &= b + c \wedge c + f = d + e \\ \Downarrow \\ a + d + f &= b + c + f = b + d + e \\ \Downarrow \\ a + f &= b + e \\ \Downarrow \\ (a, b) &\sim (e, f) \end{aligned}$$

Therefore, $\sim$ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$. The set of **integers** $\mathbb{Z}$ is defined as

$$\mathbb{Z} := (\mathbb{N} \times \mathbb{N}) / \sim$$

Definition 1.3.1. Let $[(a, b)], [(c, d)] \in \mathbb{Z}$ be given.

The **addition** $+$ on $\mathbb{Z}$ is defined as

$$[(a, b)] + [(c, d)] := [(a + c, b + d)]$$

The **multiplication** $\cdot$ on $\mathbb{Z}$ is defined as

$$[(a, b)] \cdot [(c, d)] := [(a \cdot c + b \cdot d, a \cdot d + b \cdot c)]$$

Remark. The operations $+$ and $\cdot$ are **well-defined** on $\mathbb{Z}$; that is, they do not depend on the **choice of representatives** of the equivalence classes. Indeed, if

- $(a, b) \sim (a', b')$

- $(c, d) \sim (c', d')$

then one can verify that

- $(a + c, b + d) \sim (a' + c', b' + d')$
- $(a \cdot c + b \cdot d, a \cdot d + b \cdot c) \sim (a' \cdot c' + b' \cdot d', a' \cdot d' + b' \cdot c')$

The same issue arises for **addition** and **multiplication** on $\mathbb{Q}$, whose **well-definedness** could be verified in exactly the same manner.

Theorem 1.3.2. *Let $n, m, k \in \mathbb{Z}$ be given. Then the following properties hold:*

- $n + m = m + n$
- $(n + m) + k = n + (m + k)$
- $n \cdot m = m \cdot n$
- $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- $n \cdot (m + k) = (n \cdot m) + (n \cdot k)$

Proof. The proof follows directly from [Definition 1.3.1](#) and [theorem 1.1.3](#). $\square$

Define

$$\mathbb{Z}^+ := \{[(n + 1, 1)] \in \mathbb{Z} \mid n \in \mathbb{N}\} \subseteq \mathbb{Z}$$

Define a relation $\sim$ on $\mathbb{Z} \times \mathbb{Z}^+$ as

$$(a, b) \sim (c, d) \iff a \cdot d = b \cdot c$$

Let $(a, b), (c, d), (e, f) \in \mathbb{Z} \times \mathbb{Z}^+$ be given. By [Theorem 1.3.2](#),

- **Reflexivity:** Since $a \in \mathbb{Z} \wedge b \in \mathbb{Z}^+$,

$$\begin{aligned} a \cdot b &= b \cdot a \\ \Downarrow \\ (a, b) &\sim (a, b) \end{aligned}$$

- **Symmetry:** If $(a, b) \sim (c, d)$, then

$$\begin{aligned} a \cdot d &= b \cdot c \\ \Downarrow \\ c \cdot b &= d \cdot a \\ \Downarrow \\ (c, d) &\sim (a, b) \end{aligned}$$

- **Transitivity:** If $(a, b) \sim (c, d) \wedge (c, d) \sim (e, f)$, then

$$\begin{aligned}
 a \cdot d &= b \cdot c \wedge c \cdot f = d \cdot e \\
 &\Downarrow \\
 a \cdot d \cdot f &= b \cdot c \cdot f = b \cdot d \cdot e \\
 &\Downarrow \\
 a \cdot f &= b \cdot e \\
 &\Downarrow \\
 (a, b) &\sim (e, f)
 \end{aligned}$$

Therefore, $\sim$ is an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^+$. The set of **rational numbers** $\mathbb{Q}$ is defined as

$$\mathbb{Q} := (\mathbb{Z} \times \mathbb{Z}^+) / \sim$$

Definition 1.3.3. Let $[(a, b)], [(c, d)] \in \mathbb{Q}$ be given.

The **addition** $+$ on $\mathbb{Q}$ is defined as

$$[(a, b)] + [(c, d)] := [(ad + bc, bd)]$$

The **multiplication** $\cdot$ on $\mathbb{Q}$ is defined as

$$[(a, b)] \cdot [(c, d)] := [(ac, bd)]$$

Theorem 1.3.4. Let $p, q, r \in \mathbb{Q}$ be given. Then the following properties hold:

- $p + q = q + p$
- $(p + q) + r = p + (q + r)$
- $p \cdot q = q \cdot p$
- $(p \cdot q) \cdot r = p \cdot (q \cdot r)$
- $p \cdot (q + r) = (p \cdot q) + (p \cdot r)$

Proof. The proof follows directly from [Definition 1.3.3](#) and [theorem 1.3.2](#). $\square$

Remark. Strictly speaking, the constructions of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ produce **different kind** of sets. For example,

- $n \in \mathbb{N}$
- $[(n + 1, 1)] \in \mathbb{Z}$

- $[(n, [(2, 1)])] \in \mathbb{Q}$

are distinct objects in the underlying set-theoretic construction. Nevertheless, there are **canonical embeddings**

- $\mathbb{N} \xrightarrow{\cong} \mathbb{Z}$ such that $n \mapsto [(n + 1, 1)]$
- $\mathbb{Z} \xrightarrow{\cong} \mathbb{Q}$ such that $a \mapsto [(a, [(2, 1)])]$

which preserve the **algebraic operations**. We therefore identify each number system with its image in the next larger one and write

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$$

This identification is justified by the fact that the embeddings are canonical, requiring **no arbitrary choices**. Throughout this textbook, inclusions such as

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

should be understood as canonical identifications rather than literal inclusions arising directly from the set-theoretic constructions.

Remark. Define

$$0 := [(1, 1)] \in \mathbb{Z}$$

In this textbook, we will use the following notations:

- $\mathbb{N}_0 := \mathbb{N} \cup \{0\} \subseteq \mathbb{Z}$
- $\mathbb{N}_n := \{k \in \mathbb{N} \mid k \geq n\} = \{n, n + 1, n + 2, \dots\} \subseteq \mathbb{N}$ for each $n \in \mathbb{N}$
- $\mathbb{N}^0 := \emptyset$
- $\mathbb{N}^m := \{k \in \mathbb{N} \mid k \leq m\} = \{1, 2, \dots, m\} \subseteq \mathbb{N}$ for each $m \in \mathbb{N}$
- $\mathbb{N}_n^m := \{k \in \mathbb{N}_n \mid k \leq m\} = \{n, n + 1, \dots, m\} \subseteq \mathbb{N}_0$ for each $n, m \in \mathbb{N}_0$ with $n \leq m$

1.4 Ordered Sets

A set S is called an **ordered set** if it is equipped with a relation $<$ such that for every $x, y, z \in S$, the following properties hold:

- $x < y \wedge y < z \implies x < z$
- Only one of the relations $x < y$, $x = y$, or $x > y$ holds

The relation $<$ is called an **(total) order** on S .

Definition 1.4.1. The relations $>$, $\leq$, and $\geq$ are defined as

- $x > y \iff y < x$
- $x \geq y \iff x > y \vee x = y$
- $x \leq y \iff y \geq x$

Example 1.4.2. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are ordered sets with the following relations:

- $\forall n, m \in \mathbb{N}, m < n \iff \exists k \in \mathbb{N}, n = m + k$
- $\forall n, m \in \mathbb{Z}, m < n \iff \exists k \in \mathbb{Z}^+, n = m + k$
- $\forall p, q \in \mathbb{Q}, q < p \iff \exists r \in \mathbb{Q}^+, p = q + r$

where

$$\mathbb{Q}^+ := \{[(p, q)] \in \mathbb{Q} \mid p, q \in \mathbb{Z}^+\} \subseteq \mathbb{Q}$$

Definition 1.4.3. Let S be an ordered set and $E \subseteq S$.

- E is said to be **bounded above** if

$$\exists M \in S, \forall x \in E, x \leq M$$

Such an M is called an **upper bound** of E .

- E is said to be **bounded below** if

$$\exists m \in S, \forall x \in E, m \leq x$$

Such an m is called a **lower bound** of E .

Theorem 1.4.4. Let $E \subseteq \mathbb{Z}$ be nonempty. Then

- (A) E is bounded below $\implies E$ has a least element
- (B) E is bounded above $\implies E$ has a greatest element

Proof. (A) Let E be bounded below. Then,

$$\begin{array}{c} E \text{ is bounded below} \\ \Downarrow \\ \exists m \in \mathbb{Z}, \forall x \in E, m \leq x \end{array}$$

Define

$$F := \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{aligned} E &\neq \emptyset \\ \Downarrow \\ F &\neq \emptyset \end{aligned}$$

By Theorem 1.1.4,

$$\begin{aligned} \exists n \in F, n &= \min F \\ \Downarrow \\ \forall x \in E, n &\leq x - m + 1 \\ \Downarrow \\ \forall x \in E, n + m - 1 &\leq x \end{aligned}$$

Since $n = \min F$,

$$\begin{aligned} n &\in F \\ \Downarrow \\ \exists x_0 \in E, n &= x_0 - m + 1 \\ \Downarrow \\ x_0 &= n + m - 1 = \min E \end{aligned}$$

(B) Let E be bounded above. Then,

$$\begin{aligned} E &\text{ is bounded above} \\ \Downarrow \\ \exists M \in \mathbb{Z}, \forall x \in E, x &\leq M \end{aligned}$$

Define

$$F := \{-x + M + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{aligned} E &\neq \emptyset \\ \Downarrow \\ F &\neq \emptyset \end{aligned}$$

By Theorem 1.1.4,

$$\begin{aligned} \exists n \in F, n &= \min F \\ \Downarrow \\ \forall x \in E, n &\leq -x + M + 1 \\ \Downarrow \\ \forall x \in E, x &\leq M - n + 1 \end{aligned}$$

Since $n = \min F$,

$$\begin{aligned} n &\in F \\ \Downarrow \\ \exists x_0 \in E, n &= -x_0 + M + 1 \\ \Downarrow \\ x_0 &= M - n + 1 = \max E \end{aligned}$$

□

Definition 1.4.5. Let S be an ordered set, and let $E \subseteq S$ be bounded above. An element $\alpha \in S$ is called a **supremum** (or *least upper bound*) of E if

- α is an upper bound of E
- $\forall \beta \in S, \beta < \alpha \implies \beta$ is not an upper bound of E

A supremum of E is denoted by

$$\alpha = \sup E$$

Similarly, let $E \subseteq S$ be bounded below. An element $\alpha \in S$ is called an **infimum** (or *greatest lower bound*) of E if

- α is a lower bound of E
- $\forall \beta \in S, \beta > \alpha \implies \beta$ is not a lower bound of E

An infimum of E is denoted by

$$\alpha = \inf E$$

Example 1.4.6. Define

$$S := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$$

- Let $p \in S$ be given. Define

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p$$

Then

$$\begin{aligned} q^2 - 2 &= \frac{2(p^2 - 2)}{(p + 2)^2} < 0 \\ &\Downarrow \\ q &\in S \end{aligned}$$

This implies that $p \in S$ is not an upper bound of S .

- Let $p \in \mathbb{Q}^+ \setminus S$ be given.

$$\nexists q \in \mathbb{Q}^+, q^2 = 2$$

Therefore,

$$\begin{aligned} p^2 &\geq 2 \\ &\Downarrow \\ p^2 &> 2 \\ &\Downarrow \\ \forall x \in S, x^2 &< 2 < p^2 \\ &\Downarrow \\ \forall x \in S, x &< p \\ &\Downarrow \\ p &\text{ is an upper bound of } S \end{aligned}$$

Define

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0$$

$$\Downarrow$$

$$q^2 > 2$$

$$\Downarrow$$

$$\forall x \in S, x^2 < 2 < q^2$$

$$\Downarrow$$

$$\forall x \in S, x < q$$

$$\Downarrow$$

$$q \text{ is an upper bound of } S$$

This implies that p is not a least upper bound of S .

To sum up,

$$\nexists \alpha \in \mathbb{Q}^+, \alpha = \sup S$$

Definition 1.4.7. Let S be an ordered set and $E \subseteq S$. If

- $E \neq \emptyset$
- E is bounded above

implies that

$$\exists \alpha \in S, \alpha = \sup E$$

then S is said to have the **least upper bound property**.

Theorem 1.4.8. Let S be an ordered set and $E \subseteq S$. Then

- S has the least upper bound property
- $E \neq \emptyset$
- E is bounded below

implies that

$$\exists \alpha \in S, \alpha = \inf E$$

Proof. Assume that

- S has the least upper bound property
- $E \neq \emptyset$

- E is bounded below

Define

$$L := \{\lambda \in S \mid \forall x \in E, \lambda \leq x\} \subseteq S$$

Then

E is bounded below

$\Downarrow$

$$L \neq \emptyset$$

Also,

$$E \neq \emptyset$$

$\Downarrow$

$$\exists \beta \in E$$

$\Downarrow$

$$\forall \lambda \in L, \lambda \leq \beta$$

$\Downarrow$

L is bounded above

By the *least upper bound property*,

$$\exists \alpha \in S, \alpha = \sup L$$

We will show that

$$\alpha = \inf E$$

Assume that $\alpha \notin L$. Then

$$\alpha \notin L$$

$\Downarrow$

$$\exists \beta \in E, \beta < \alpha$$

Since $\alpha = \sup L$,

$$\alpha = \sup L$$

$\Downarrow$

β is not an upper bound of L

$\Downarrow$

$$\exists \gamma \in L, \beta < \gamma \leq \alpha$$

Then

$$\gamma \in L$$

$\Downarrow$

$$\forall x \in E, \gamma \leq x$$

$\Downarrow$

$$\gamma \leq \beta$$

$\Downarrow$

$$\beta < \gamma \leq \beta$$

$\Downarrow$

$\perp$

Therefore,

$$\begin{array}{c} \alpha \in L \\ \Downarrow \\ \alpha \text{ is a lower bound of } E \end{array}$$

Since $\alpha = \sup L$,

$$\forall \delta \in S, \delta > \alpha \implies \delta \notin L$$

To sum up,

- α is a lower bound of E
- $\forall \delta \in S, \delta > \alpha \implies \delta$ is not a lower bound of E

Therefore,

$$\alpha = \inf E$$

□

1.5 Fields and Ordered Fields

A set F is called a **field** if it is equipped with two binary operations

- **addition** $+$
- **multiplication** $\cdot$

and satisfies the following axioms:

(A) Axioms for **addition**

- $\forall x, y \in F, x + y \in F$
- $\forall x, y \in F, x + y = y + x$
- $\forall x, y, z \in F, (x + y) + z = x + (y + z)$
- $\exists 0 \in F, \forall x \in F, x + 0 = x$
- $\forall x \in F, \exists -x \in F, x + (-x) = 0$

(B) Axioms for **multiplication**

- $\forall x, y \in F, x \cdot y \in F$
- $\forall x, y \in F, x \cdot y = y \cdot x$
- $\forall x, y, z \in F, (x \cdot y) \cdot z = x \cdot (y \cdot z)$
- $\exists 1 \in F, 1 \neq 0 \wedge (\forall x \in F, x \cdot 1 = x)$
- $\forall x \in F, x \neq 0 \implies \exists x^{-1} \in F, x \cdot x^{-1} = 1$

(C) The **distributive law**

$$\bullet \forall x, y, z \in F, x \cdot (y + z) = x \cdot y + x \cdot z$$

A set F is called an **ordered field** if

- F is a field
- F is an ordered set

and satisfies the following axioms:

- $\forall x, y, z \in F, x < y \implies x + z < y + z$
- $\forall x, y \in F, x > 0 \wedge y > 0 \implies x \cdot y > 0$

Example 1.5.1. $(\mathbb{Q}, +, \cdot)$ is an ordered field, while $(\mathbb{Z}, +, \cdot)$ is not an ordered field since

$$\nexists 2^{-1} \in \mathbb{Z}, 2 \cdot 2^{-1} = 1$$

1.6 The Real Number System

An ordered field $(\mathbb{R}, +, \cdot)$ has the *least upper bound property*. The construction of $\mathbb{R}$ from $\mathbb{Q}$ is rather complicated, so it will be discussed separately in [Appendix](#). In this section, we will just assume that $\mathbb{R}$ exists.

Theorem 1.6.1. $\mathbb{R}$ has the *Archimedean property*, which states that

$$\forall x \in \mathbb{R}^+, \forall y \in \mathbb{R}, \exists n \in \mathbb{N}, nx > y$$

where

$$\mathbb{R}^+ := \{x \in \mathbb{R} \mid x > 0\}$$

Proof. Define

$$E := \{nx \mid nx \leq y, n \in \mathbb{N}\} \subseteq \mathbb{R}$$

If $E = \emptyset$, then

$$\begin{aligned} 1 \in \mathbb{N} \wedge 1 \cdot x \notin E \\ \Downarrow \\ 1 \cdot x > y \end{aligned}$$

Otherwise,

$$E \neq \emptyset$$

E is bounded above by y . By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}, \alpha = \sup E$$

Since $\alpha = \sup E$,

$$\begin{aligned}
 & \alpha - x < \alpha \\
 & \Downarrow \\
 & \alpha - x \text{ is not an upper bound of } E \\
 & \Downarrow \\
 & \exists m \in \mathbb{N}, mx > \alpha - x \\
 & \Downarrow \\
 & (m+1)x > \alpha \\
 & \Downarrow \\
 & m+1 \in \mathbb{N} \wedge (m+1)x \notin E \\
 & \Downarrow \\
 & (m+1)x > y
 \end{aligned}$$

Therefore, we can take $n = m + 1$ to complete the proof. $\square$

Corollary 1.6.2.

$$\forall \varepsilon \in \mathbb{R}^+, \forall b \in (1, +\infty), \exists n \in \mathbb{N}, b^{-n} < \varepsilon$$

Proof. The **Bernoulli's inequality** states that

$$\forall x \in [-1, +\infty), \forall n \in \mathbb{N}, (1+x)^n \geq 1+nx$$

By [Theorem 1.6.1](#),

$$\begin{aligned}
 & \forall a \in \mathbb{R}^+, \exists n \in \mathbb{N}, na > \frac{1}{\varepsilon} \\
 & \Downarrow \\
 & (1+a)^n \geq 1+na > \frac{1}{\varepsilon} \implies (1+a)^{-n} < \varepsilon
 \end{aligned}$$

Then

$$\begin{aligned}
 & b \in (1, +\infty) \\
 & \Downarrow \\
 & b-1 \in \mathbb{R}^+ \\
 & \Downarrow \\
 & \exists n \in \mathbb{N}, n(b-1) > \frac{1}{\varepsilon} \\
 & \Downarrow \\
 & (1+(b-1))^n \geq 1+n(b-1) > \frac{1}{\varepsilon} \implies (1+(b-1))^{-n} < \varepsilon \\
 & \Downarrow \\
 & b^{-n} < \varepsilon
 \end{aligned}$$

$\square$

Theorem 1.6.3. *Let $E \subseteq \mathbb{Z}$ be nonempty. Then*

(A) *E is bounded below in $\mathbb{R} \implies E$ has a least element*

(B) *E is bounded above in $\mathbb{R} \implies E$ has a greatest element*

Proof. (A) Let E be bounded below in $\mathbb{R}$. Then

$$\exists \alpha \in \mathbb{R}, \forall x \in E, \alpha \leq x$$

If $\alpha \geq 0$, then

$$\forall x \in E, 0 \leq \alpha \leq x$$

$$\Downarrow$$

$0 \in \mathbb{Z}$ is a lower bound of E

Otherwise, by [Theorem 1.6.1](#),

$$-\alpha \in \mathbb{R}$$

$$\Downarrow$$

$$\exists n \in \mathbb{N}, n > -\alpha$$

$$\Downarrow$$

$$\forall x \in E, -n < \alpha \leq x$$

$$\Downarrow$$

$-n \in \mathbb{Z}$ is a lower bound of E

Therefore, E is bounded below in $\mathbb{Z}$. By [Theorem 1.4.4](#), E has a least element.

(B) Let E be bounded above in $\mathbb{R}$. Then

$$\exists \alpha \in \mathbb{R}, \forall x \in E, x \leq \alpha$$

Then

$$\alpha \in \mathbb{R}$$

$$\Downarrow$$

$$\exists n \in \mathbb{N}, n > \alpha$$

$$\Downarrow$$

$$\forall x \in E, x \leq \alpha < n$$

$$\Downarrow$$

$n \in \mathbb{Z}$ is an upper bound of E

Therefore, E is bounded above in $\mathbb{Z}$. By [Theorem 1.4.4](#), E has a greatest element.

□

Definition 1.6.4. Let $x \in \mathbb{R}$ be given. The **floor function** $\lfloor x \rfloor$ and **ceiling function** $\lceil x \rceil$ are defined as

$$\bullet \lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$$

$$\bullet \lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$$

[Theorem 1.6.3](#) guarantees the *existence* of $\lfloor x \rfloor$ and $\lceil x \rceil$.

Theorem 1.6.5. $\mathbb{Q}$ is *dense* in $\mathbb{R}$, which states that

$$\forall x, y \in \mathbb{R}, x < y \implies \exists p \in \mathbb{Q}, x < p < y$$

Proof. Define

$$\varepsilon := y - x > 0$$

By [Theorem 1.6.1](#),

$$\exists n \in \mathbb{N}, \frac{1}{n} < \varepsilon$$

Define

$$S := \left\{ k \in \mathbb{Z} \mid \frac{k}{n} \leq x \right\} \subseteq \mathbb{Z}$$

Then

$$\begin{aligned} \lfloor x \rfloor &\leq x < \lfloor x \rfloor + 1 \\ &\Downarrow \\ n\lfloor x \rfloor &\leq nx < n\lfloor x \rfloor + n \\ &\Downarrow \\ n\lfloor x \rfloor &\in S \\ &\Downarrow \\ S &\neq \emptyset \end{aligned}$$

S is bounded above by nx . By [Theorem 1.6.3](#),

$$\exists m \in \mathbb{Z}, m = \max S$$

Since $m = \max S$,

$$\begin{aligned} m &\in S \\ &\Downarrow \\ \frac{m}{n} &\leq x \\ &\Downarrow \\ \frac{m+1}{n} &\leq x + \frac{1}{n} \end{aligned}$$

Also,

$$\begin{aligned} m+1 &\notin S \\ &\Downarrow \\ \frac{m+1}{n} &> x \end{aligned}$$

To sum up,

$$x < \frac{m+1}{n} \leq x + \frac{1}{n} < x + \varepsilon = x + (y - x) = y$$

Therefore, we can take $p = \frac{m+1}{n} \in \mathbb{Q}$ to complete the proof. □

Theorem 1.6.6.

$$\forall x \in \mathbb{R}^+, \forall n \in \mathbb{N}, \exists! y \in \mathbb{R}^+, y^n = x$$

Such y is denoted by

- $y = \sqrt[n]{x}$
- $y = x^{\frac{1}{n}}$

Proof. Define

$$E := \{t \in \mathbb{R}^+ \mid t^n < x\}$$

If $x \geq 1$, then

$$\forall m \in \mathbb{N}, 2^{-m} < 1 \leq x$$

$$\Downarrow$$

$$2^{-1} \in E$$

$$\Downarrow$$

$$E \neq \emptyset$$

Otherwise,

$$x < 1$$

$$\Downarrow$$

$$\forall m \in \mathbb{N}, x^{2m} < x$$

$$\Downarrow$$

$$x^2 \in E$$

$$\Downarrow$$

$$E \neq \emptyset$$

E is bounded above by $\max\{x, 1\}$. By the *least upper bound property*,

$$\exists \alpha \in \mathbb{R}^+, \alpha = \sup E$$

We will show that $\alpha^n = x$. Let $0 < a < b < \infty$ be given. Then

$$b^n - a^n = (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k < (b - a) n b^{n-1}$$

Assume that $\alpha^n < x$. By [Theorem 1.6.5](#),

$$\exists h \in \mathbb{R}^+, h < \min \left\{ 1, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}} \right\}$$

Put $a = \alpha$ and $b = \alpha + h$. Then

$$(\alpha + h)^n - \alpha^n < hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1}$$

$$\Downarrow$$

$$(\alpha + h)^n < \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x$$

Since $\alpha = \sup E$,

$$\begin{aligned} (\alpha + h)^n &< x \\ \Downarrow \\ \alpha + h &\in E \\ \Downarrow \\ &\perp \end{aligned}$$

Assume that $\alpha^n > x$. By [Theorem 1.6.5](#),

$$\exists h \in \mathbb{R}^+, h < \min \left\{ \alpha, \frac{\alpha^n - x}{n\alpha^{n-1}} \right\}$$

Put $a = \alpha - h$ and $b = \alpha$. Then

$$\begin{aligned} \alpha^n - (\alpha - h)^n &< hn\alpha^{n-1} \\ \Downarrow \\ (\alpha - h)^n &> \alpha^n - hn\alpha^{n-1} > \alpha^n - (\alpha^n - x) = x \end{aligned}$$

Since $\alpha = \sup E$,

$$\begin{aligned} (\alpha - h)^n &> x \\ \Downarrow \\ \forall t \in E, t^n &< x < (\alpha - h)^n \\ \Downarrow \\ \forall t \in E, t &< \alpha - h \\ \Downarrow \\ \alpha - h &\text{ is an upper bound of } E \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\alpha^n = x$$

This guarantees the *existence* of y . Let $0 < \alpha < \beta < \infty$ be given. Then

$$\begin{aligned} \beta^n - \alpha^n &= (\beta - \alpha) \sum_{k=0}^{n-1} \beta^{n-1-k} \alpha^k > 0 \\ \Downarrow \\ \beta^n &> \alpha^n \end{aligned}$$

This guarantees the *uniqueness* of y . □

Definition 1.6.7. Let $y \in [0, 1)$ and $b \in \mathbb{N}_2$ be given. A sequence $(a_n)_{n=1}^{\infty}$ is called a **greedy base b expansion** of y if

$$\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$$

where

$$\forall k \in \mathbb{N}, a_k \in S_b := \{0, 1, 2, \dots, b-1\}$$

In particular, if $b = 10$, such a sequence is called a **greedy decimal expansion** of y .

Theorem 1.6.8. *Let $y \in [0, 1)$ and $b \in \mathbb{N}_2$ be given. Then the greedy base b expansion of y exists and is unique.*

Proof. Define

$$\begin{array}{lll} t_1 := by & T_1 := \{t \in S_b \mid t \leq t_1\} & a_1 := \max T_1 \\ t_2 := b^2y - ba_1 & T_2 := \{t \in S_b \mid t \leq t_2\} & a_2 := \max T_2 \\ \vdots & \vdots & \vdots \\ t_n := b^ny - \sum_{k=1}^{n-1} a_k b^{n-k} & T_n := \{t \in S_b \mid t \leq t_n\} & a_n := \max T_n \end{array}$$

Let $m \in \mathbb{N}$ be given. If $0 \leq t_m < b$, then

- $0 \in T_m$
- T_m is bounded above by $b-1$

By [Theorem 1.4.4](#),

$$\begin{array}{c} \exists M \in S_b, M = \max T_m \\ \Downarrow \\ a_m = M \text{ is well-defined} \end{array}$$

Define $P(n)$:

- $0 \leq t_n < b$
- $\forall k \in \mathbb{N}^n, a_k = \max T_k$ is well-defined

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Since $0 \leq t_1 = by < b$,

- $0 \in T_1$
- T_1 is bounded above by $b-1$

By [Theorem 1.4.4](#),

$$\begin{array}{c} a_1 = \max T_1 \text{ is well-defined} \\ \Downarrow \\ P(1) \text{ is true} \end{array}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Then

$$\begin{aligned}
 t_{n+1} &= b^{n+1}y - \sum_{k=1}^n a_k b^{n+1-k} \\
 &= b \left(b^n y - \sum_{k=1}^n a_k b^{n-k} \right) \\
 &= b \left(b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n \right) \\
 &= b(t_n - a_n)
 \end{aligned}$$

By the inductive hypothesis,

- $0 \leq t_n < b$
- $a_n = \max T_n$

Then

$$\begin{aligned}
 a_n &\leq t_n < a_n + 1 \\
 &\Downarrow \\
 0 &\leq t_n - a_n < 1 \\
 &\Downarrow \\
 0 &\leq t_{n+1} = b(t_n - a_n) < b \\
 &\Downarrow \\
 P(n+1) &\text{ is true}
 \end{aligned}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. Then

$$\begin{aligned}
 a_n &\leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\
 &\Downarrow \\
 0 &\leq b^n y - \sum_{k=1}^n a_k b^{n-k} < 1 \\
 &\Downarrow \\
 0 &\leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}
 \end{aligned}$$

for all $n \in \mathbb{N}$. This guarantees the *existence* of a *greedy base b expansion* of y . Let two sequences $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ satisfy

- $\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$
- $\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n}$

where

$$\forall k \in \mathbb{N}, a_k \in S_b \wedge c_k \in S_b$$

We will show that

$$\forall n \in \mathbb{N}, a_n = c_n$$

Define

$$Q(n) : \forall k \in \mathbb{N}^n, a_k = c_k$$

We will show that $Q(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** By the assumption,

$$\begin{array}{ccc} 0 \leq y - \frac{a_1}{b} < \frac{1}{b} & & 0 \leq y - \frac{c_1}{b} < \frac{1}{b} \\ \Downarrow & & \Downarrow \\ \frac{a_1}{b} \leq y < \frac{a_1 + 1}{b} & & \frac{c_1}{b} \leq y < \frac{c_1 + 1}{b} \end{array}$$

By combining the two inequalities,

$$\begin{array}{ccc} \frac{a_1}{b} \leq y < \frac{c_1 + 1}{b} & & \frac{c_1}{b} \leq y < \frac{a_1 + 1}{b} \\ \Downarrow & & \Downarrow \\ a_1 < c_1 + 1 & & c_1 < a_1 + 1 \\ \Downarrow & & \Downarrow \\ a_1 \leq c_1 & & c_1 \leq a_1 \end{array}$$

Therefore,

$$\begin{array}{c} a_1 = c_1 \\ \Downarrow \\ Q(1) \text{ is true} \end{array}$$

- **Inductive step:** Let $Q(n)$ be true for $n \in \mathbb{N}$. Then

$$\begin{array}{ccc} 0 \leq y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} & & 0 \leq y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow & & \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} & & \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \end{array}$$

By combining the two inequalities,

$$\begin{array}{ccc} \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} & & \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow & & \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}} & & \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}} \end{array}$$

By the inductive hypothesis,

$$\forall k \in \mathbb{N}^n, a_k = c_k$$

Then

$$\begin{array}{ccc} \frac{a_{n+1} - c_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} & & \frac{c_{n+1} - a_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} \\ \Downarrow & & \Downarrow \\ a_{n+1} - c_{n+1} < 1 & & c_{n+1} - a_{n+1} < 1 \\ \Downarrow & & \Downarrow \\ a_{n+1} \leq c_{n+1} & & c_{n+1} \leq a_{n+1} \end{array}$$

Therefore,

$$\begin{array}{c} a_{n+1} = c_{n+1} \\ \Downarrow \\ Q(n+1) \text{ is true} \end{array}$$

By the *principle of mathematical induction*, $Q(n)$ is true for all $n \in \mathbb{N}$. This guarantees the *uniqueness* of the greedy base b expansion of y . $\square$

Theorem 1.6.9. Let $b \in \mathbb{N}_2$ be given. Let

- $(a_n)_{n=1}^\infty$ be the greedy base b expansion of $a \in [0, 1)$
- $(c_n)_{n=1}^\infty$ be the greedy base b expansion of $c \in [0, 1)$

Then

$$(\forall n \in \mathbb{N}, a_n = c_n) \implies a = c$$

Proof. By [Corollary 1.6.2](#), if $a \neq c$ (WLOG $a < c$), then

$$\exists n \in \mathbb{N}, \frac{1}{b^n} < c - a$$

By [Definition 1.6.7](#),

$$\begin{array}{ccc} 0 \leq a - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} & & 0 \leq c - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n} \\ \Downarrow & & \Downarrow \\ \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} & & c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n} \end{array}$$

By the assumption,

$$\begin{aligned}
 & \forall n \in \mathbb{N}, a_n = c_n \\
 & \Downarrow \\
 & \sum_{k=1}^n \frac{a_k}{b^k} = \sum_{k=1}^n \frac{c_k}{b^k} \\
 & \Downarrow \\
 & \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n} \\
 & \Downarrow \\
 & \perp
 \end{aligned}$$

Therefore,

$$a = c$$

□

Theorem 1.6.10. *Let $b \in \mathbb{N}_2$ be given. Let $(a_n)_{n=1}^\infty$ be such that*

$$\forall k \in \mathbb{N}, a_k \in S_b$$

If

$$\nexists y \in [0, 1), (a_n)_{n=1}^\infty \text{ is the greedy base } b \text{ expansion of } y$$

then

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

Proof. We will prove the contrapositive: If

$$\forall N \in \mathbb{N}, \exists n \in \mathbb{N}_N, a_n \neq b - 1$$

then

$$\exists y \in [0, 1), (a_n)_{n=1}^\infty \text{ is the greedy base } b \text{ expansion of } y$$

Assume that

$$\forall N \in \mathbb{N}, \exists n \in \mathbb{N}_N, a_n \neq b - 1$$

Define

$$x_n := \sum_{k=1}^n \frac{a_k}{b^k}$$

for each $n \in \mathbb{N}$. Then

$$\begin{aligned}
 & \forall k \in \mathbb{N}, a_k \in S_b \\
 & \Downarrow \\
 & x_n = \sum_{k=1}^n \frac{a_k}{b^k} \leq \sum_{k=1}^n \frac{b-1}{b^k} = 1 - \frac{1}{b^n} < 1
 \end{aligned}$$

for all $n \in \mathbb{N}$. Therefore, $\{x_n\}_{n=1}^{\infty}$ is bounded above by 1. By the *least upper bound property*,

$$\exists y \in \mathbb{R}, y = \sup\{x_n\}_{n=1}^{\infty}$$

First, we will show that $y \in [0, 1)$. By the assumption,

$$\exists m \in \mathbb{N}, a_m \neq b - 1$$

Then

$$\begin{aligned} x_n &= x_m + \sum_{k=m+1}^n \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^n \frac{b-1}{b^k} \\ &\Downarrow \\ x_n &\leq x_m + \frac{1}{b^m} - \frac{1}{b^n} < x_m + \frac{1}{b^m} \end{aligned}$$

for all $n > m$. Since $a_m \leq b - 2$,

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^m \frac{b-1}{b^k} = 1 - \frac{1}{b^m} \end{aligned}$$

To sum up,

- $\forall n \in \mathbb{N}, n \leq m \implies x_n \leq x_m$
- $\forall n \in \mathbb{N}, n > m \implies x_n < x_m + \frac{1}{b^m} \leq 1 - \frac{1}{b^m}$

Therefore, $1 - \frac{1}{b^m}$ is an upper bound of $\{x_n\}_{n=1}^{\infty}$. Since $0 \leq x_1 \leq y$,

$$\begin{aligned} y &= \sup\{x_n\}_{n=1}^{\infty} \\ &\Downarrow \\ y &\leq 1 - \frac{1}{b^m} < 1 \\ &\Downarrow \\ y &\in [0, 1) \end{aligned}$$

Next, we will show that $(a_n)_{n=1}^{\infty}$ is the *greedy base b expansion* of y . Let $n \in \mathbb{N}$ be fixed. By the assumption,

$$\exists m \in \mathbb{N}_{n+1}, a_m \leq b - 2$$

Then

$$\begin{aligned}
 x_m &= x_n + \sum_{k=n+1}^m \frac{a_k}{b^k} < x_n + \sum_{k=n+1}^m \frac{b-1}{b^k} \\
 &\Downarrow \\
 x_m &< x_n + \frac{1}{b^n} - \frac{1}{b^m} \\
 &\Downarrow \\
 x_m + \frac{1}{b^m} &< x_n + \frac{1}{b^n}
 \end{aligned}$$

By the same argument as above, $x_m + \frac{1}{b^m}$ is an upper bound of $\{x_n\}_{n=1}^\infty$. Then

$$\begin{aligned}
 y &= \sup\{x_n\}_{n=1}^\infty \\
 &\Downarrow \\
 y &\leq x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n} \\
 &\Downarrow \\
 0 &\leq y - x_n < \frac{1}{b^n}
 \end{aligned}$$

Since n was arbitrary, $(a_n)_{n=1}^\infty$ is the greedy base b expansion of y . □

Theorem 1.6.11. Let $b \in \mathbb{N}_2$ be given. Let $(a_n)_{n=1}^\infty$ be such that

$$\forall k \in \mathbb{N}, a_k \in S_b$$

If

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

then

$$\nexists y \in [0, 1), (a_n)_{n=1}^\infty \text{ is the greedy base } b \text{ expansion of } y$$

This theorem is the **converse** of [Theorem 1.6.10](#).

Proof. Assume that

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = b - 1$$

If

$$\exists y \in [0, 1), (a_n)_{n=1}^\infty \text{ is the greedy base } b \text{ expansion of } y$$

then

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$$

for all $n \in \mathbb{N}$. Define

$$x_n := \sum_{k=1}^n \frac{a_k}{b^k}$$

for each $n \in \mathbb{N}$. Then

$$\begin{aligned}
 x_n &= x_N + \sum_{k=N+1}^n \frac{b-1}{b^k} = x_N + \frac{1}{b^N} - \frac{1}{b^n} \\
 &\Downarrow \\
 y - x_n &= y - x_N - \frac{1}{b^N} + \frac{1}{b^n} \geq 0 \\
 &\Downarrow \\
 \frac{1}{b^n} &\geq x_N + \frac{1}{b^N} - y
 \end{aligned}$$

for all $n > N$. Since $N \in \mathbb{N}$,

$$\begin{aligned}
 0 &\leq y - x_N < \frac{1}{b^N} \\
 &\Downarrow \\
 x_N + \frac{1}{b^N} - y &> 0
 \end{aligned}$$

By [Corollary 1.6.2](#),

$$\begin{aligned}
 \exists m \in \mathbb{N}, \frac{1}{b^m} &< x_N + \frac{1}{b^N} - y \\
 &\Downarrow \\
 \forall n \in \mathbb{N}, n > \max\{m, N\} &\implies \frac{1}{b^n} < x_N + \frac{1}{b^N} - y \\
 &\Downarrow \\
 &\perp
 \end{aligned}$$

Therefore,

$$\nexists y \in [0, 1), (a_n)_{n=1}^\infty \text{ is the greedy base } b \text{ expansion of } y$$

□

1.7 The Extended Real Number System

The Extended real number system $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$$

We define the following operations on $\overline{\mathbb{R}}$:

- $\forall x \in \mathbb{R} \cup \{+\infty\}, x + (+\infty) := +\infty$
- $\forall x \in \mathbb{R} \cup \{-\infty\}, x + (-\infty) := -\infty$
- $\forall x \in \mathbb{R}^+ \cup \{+\infty\}, x \cdot (\pm\infty) := \pm\infty$
- $\forall x \in \mathbb{R}^- \cup \{-\infty\}, x \cdot (\pm\infty) := \mp\infty$

- $\forall x \in \mathbb{R}, \frac{x}{\pm\infty} := 0$

The following operations are undefined in $\overline{\mathbb{R}}$:

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\frac{(\pm\infty)}{(\pm\infty)}$ and $\frac{(\pm\infty)}{(\mp\infty)}$
- $\frac{x}{0}$ for all $x \in \overline{\mathbb{R}}$

$\overline{\mathbb{R}}$ is not a field, but it is an ordered set with the following order relation:

$$\forall x \in \mathbb{R}, -\infty < x < +\infty$$

Remark. We can identify the following intervals in $\overline{\mathbb{R}}$:

- $\forall a \in \mathbb{R}, (-\infty, a) = \{x \in \mathbb{R} \mid x < a\}$
- $\forall a \in \mathbb{R}, (a, +\infty) = \{x \in \mathbb{R} \mid x > a\}$
- $(-\infty, +\infty) = \mathbb{R}$.

Theorem 1.7.1. For all $E \subseteq \overline{\mathbb{R}}$,

- $\exists x \in \overline{\mathbb{R}}, x = \sup E$
- $\exists y \in \overline{\mathbb{R}}, y = \inf E$

Proof. If $E = \emptyset$, then

- $\sup E = -\infty$
- $\inf E = +\infty$

Otherwise,

- If E is bounded above in $\mathbb{R}$, then

$$\exists x \in \mathbb{R}, x = \sup E$$

by the *least upper bound property*.

- Otherwise, $\sup E = +\infty$

Also,

- If E is bounded below in $\mathbb{R}$, then

$$\exists y \in \mathbb{R}, y = \inf E$$

by [Theorem 1.4.8](#).

- Otherwise, $\inf E = -\infty$

□

1.8 The Complex Number System

The complex number system $\mathbb{C}$ is defined as

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}$$

The binary operations on $\mathbb{C}$ are defined as

- $\forall x, y, u, v \in \mathbb{R}, (x, y) + (u, v) := (x + u, y + v)$
- $\forall x, y, u, v \in \mathbb{R}, (x, y) \cdot (u, v) := (xu - yv, xv + yu)$

One can verify that $\mathbb{C}$ is a field with the following properties:

- The **additive identity** is $(0, 0)$
- The **multiplicative identity** is $(1, 0)$
- The **additive inverse** of (x, y) is $(-x, -y)$
- The **multiplicative inverse** of (x, y) is

$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right)$$

for $(x, y) \neq (0, 0)$

The **imaginary unit** i is defined as

$$i := (0, 1) \in \mathbb{C}$$

Then

$$i^2 = (0, 1) \cdot (0, 1) = (-1, 0)$$

There is a **canonical embedding**

$$\mathbb{R} \xrightarrow{\cong} \mathbb{C} \text{ such that } x \mapsto (x, 0)$$

Therefore, we identify $\mathbb{R}$ with its image in $\mathbb{C}$ and write

$$\mathbb{R} \subseteq \mathbb{C}$$

Then, for every complex number $(x, y) \in \mathbb{C}$,

$$(x, y) = (x, 0) + (0, y) = (x, 0) + (y, 0) \cdot (0, 1) = x + yi$$

Definition 1.8.1. Let $z = (x, y) \in \mathbb{C}$ be given.

- The **real part** of z is defined as

- $\Re(z) := x$
- $\operatorname{Re}(z) := x$
- The **imaginary part** of z is defined as
 - $\Im(z) := y$
 - $\operatorname{Im}(z) := y$
- The **complex conjugate** of z is defined as

$$\bar{z} := (x, -y) = x - yi$$

Theorem 1.8.2. *Let $z = (x, y) \in \mathbb{C}$ and $w = (u, v) \in \mathbb{C}$ be given. Then the following properties hold:*

- $\overline{(\bar{z})} = z$
- $z = \bar{z} \iff z \in \mathbb{R}$
- $\overline{z + w} = \bar{z} + \bar{w}$
- $\overline{zw} = \bar{z} \cdot \bar{w}$
- $z + \bar{z} = 2\operatorname{Re}(z)$
- $z - \bar{z} = 2i\operatorname{Im}(z)$
- $z\bar{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$

Proof. The proof follows directly from [Definition 1.8.1](#). □

Definition 1.8.3. Let $z = (x, y) \in \mathbb{C}$ be given. The **modulus** of z is defined as

$$|z| := \sqrt{z\bar{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}$$

The *existence* and *uniqueness* of $|z|$ are guaranteed by [Theorems 1.6.6](#) and [1.8.2](#)

Theorem 1.8.4. *Let $a_1, a_2, \dots, a_n \in \mathbb{C}$ and $b_1, b_2, \dots, b_n \in \mathbb{C}$ be given. Then*

$$\left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 \leq \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2$$

*This theorem is called the **Cauchy-Schwarz inequality**.*

Proof. Let $1 \leq i < j \leq n$ be given. Then

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i) \overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \bar{b}_i \cdot \bar{a}_j b_j + \bar{a}_i b_i \cdot a_j \bar{b}_j) \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i < j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i < j} (|a_i|^2 |b_j|^2) + \sum_{j < i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2 \end{aligned}$$

$$\begin{aligned} \sum_{i < j} (a_i \bar{b}_i \cdot \bar{a}_j b_j + \bar{a}_i b_i \cdot a_j \bar{b}_j) &= \sum_{i \neq j} (a_i \bar{b}_i \cdot \bar{a}_j b_j) \\ &= \sum_{i=1}^n a_i \bar{b}_i \cdot \sum_{j=1}^n \bar{a}_j b_j - \sum_{k=1}^n a_k \bar{b}_k \cdot \bar{a}_k b_k \\ &= \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2 \end{aligned}$$

Therefore,

$$\sum_{i < j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 \geq 0$$

□

Corollary 1.8.5. Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ and $b_1, b_2, \dots, b_n \in \mathbb{R}$ be given. Then

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \sum_{k=1}^n a_k^2 \cdot \sum_{k=1}^n b_k^2$$

Theorem 1.8.6. Let $z, w \in \mathbb{C}$ be given. Then the following properties hold:

(A) $|zw| = |z| \cdot |w|$

(B) $|z + w| \leq |z| + |w|$

(C) $||z| - |w|| \leq |z - w|$

Proof. Let $z = (x, y) \in \mathbb{C}$ and $w = (u, v) \in \mathbb{C}$ be given.

$$(A) \quad |zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$$

(B) By [Corollary 1.8.5](#),

$$(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$$

Then

$$\begin{aligned} |z + w|^2 &= (x + u)^2 + (y + v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2 \end{aligned}$$

(C) Again, by [Corollary 1.8.5](#),

$$\begin{aligned} |z - w|^2 &= (x - u)^2 + (y - v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2 \end{aligned}$$

□

1.9 Euclidean Spaces

For each $n \in \mathbb{N}$, the **Euclidean n -space** $\mathbb{R}^n$ is defined as

$$\mathbb{R}^n := \{\mathbf{x} = (x_1, x_2, \dots, x_n) \mid \forall k \in \mathbb{N}^n, x_k \in \mathbb{R}\}$$

where $x_1, x_2, \dots, x_n$ are called the **coordinates** of $\mathbf{x}$. The elements of $\mathbb{R}^n$ for $n > 1$ are called **points** (or *vectors*) and are represented by bold symbols.

The operations on $\mathbb{R}^n$ are defined as

- $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{x} + \mathbf{y} := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$
- $\forall \alpha \in \mathbb{R}, \forall \mathbf{x} \in \mathbb{R}^n, \alpha \mathbf{x} := (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$

Definition 1.9.1. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ be given.

- The **inner product** of $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$\mathbf{x} \cdot \mathbf{y} := \sum_{k=1}^n x_k y_k$$

- The **norm** of $\mathbf{x}$ is defined as

$$\|\mathbf{x}\| := \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{k=1}^n x_k^2}$$

- The **distance** between $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

where one can verify that $d(x, y) = |x - y|$ in the case of $n = 1$.

Theorem 1.9.2. *Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ be given. Then the following properties hold:*

- (A) $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$
- (B) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$
- (C) $\|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x}\| - \|\mathbf{y}\|$

Proof. (A) By [Corollary 1.8.5](#),

$$|\mathbf{x} \cdot \mathbf{y}|^2 = \left| \sum_{k=1}^n x_k y_k \right|^2 \leq \sum_{k=1}^n x_k^2 \cdot \sum_{k=1}^n y_k^2 = \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2$$

(B) By part (A),

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2|\mathbf{x} \cdot \mathbf{y}| + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \cdot \|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2 \end{aligned}$$

(C) By part (B),

$$\begin{aligned} \|\mathbf{x}\| &= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| \\ &\Downarrow \\ \|\mathbf{x}\| - \|\mathbf{y}\| &\leq \|\mathbf{x} - \mathbf{y}\| \end{aligned}$$

Also,

$$\begin{aligned} \|\mathbf{y}\| &= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| \\ &\Downarrow \\ \|\mathbf{y}\| - \|\mathbf{x}\| &\leq \|\mathbf{x} - \mathbf{y}\| \end{aligned}$$

To sum up,

$$\|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x}\| - \|\mathbf{y}\|$$

□

Chapter 2

Basic Topology

2.1 Functions

A **function** (or *mapping*) is defined with the following three components:

- A **domain** X
- A **codomain** Y
- A **rule** f which assigns a *unique* element $y \in Y$ to each $x \in X$

f is said to be a function from X to Y , which is denoted by

$$f : X \rightarrow Y$$

For each $x \in X$, the *unique* element $y \in Y$ assigned to x is denoted by

$$y = f(x)$$

Definition 2.1.1. Let $f : X \rightarrow Y$ be given. For every $E \subseteq X$, the **image** of E under f is defined as

$$f(E) := \{f(x) \mid x \in E\} \subseteq Y$$

In particular, if $E = X$, then $f(X) \subseteq Y$ is called the **range** of f .

Definition 2.1.2. Let $f : X \rightarrow Y$ be given. For every $E \subseteq Y$, the **inverse image** of E under f is defined as

$$f^{-1}(E) := \{x \in X \mid f(x) \in E\} \subseteq X$$

Definition 2.1.3. Let $f : X \rightarrow Y$ be given.

- f is **injective** (or *one-to-one*) if

$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$$

- f is **surjective** (or *onto*) if

$$\forall y \in Y, \exists x \in X, f(x) = y$$

or equivalently,

$$f(X) = Y$$

- f is **bijective** if it is both injective and surjective

Remark. The following notations are used to denote the properties of functions:

- $f : X \hookrightarrow Y$ denotes that f is injective
- $f : X \twoheadrightarrow Y$ denotes that f is surjective
- $f : X \xrightarrow{\sim} Y$ or $f : X \xrightarrow{\cong} Y$ denotes that f is bijective

Definition 2.1.4. Let $f : X \xrightarrow{\sim} Y$ be given. The **inverse function** $f^{-1} : Y \rightarrow X$ of f is defined as

$$\forall x \in X, f^{-1}(f(x)) = x$$

If

- $f^{-1}(y) = x_1$
- $f^{-1}(y) = x_2$

then

- $f(x_1) = y$
- $f(x_2) = y$

Since f is injective,

$$\begin{array}{c} f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2 \end{array}$$

Since f is surjective,

$$\forall y \in Y, \exists x \in X, f(x) = y$$

Therefore, f^{-1} is well-defined.

Theorem 2.1.5. *Let $f : X \xrightarrow{\sim} Y$ be given. Then $f^{-1} : Y \rightarrow X$ is bijective.*

Proof. Let $x \in X$ and $y_1, y_2 \in Y$ be given.

- **Injectivity:** If

$$f^{-1}(y_1) = f^{-1}(y_2) = x$$

then, by [Definition 2.1.4](#),

- $f(x) = y_1$
- $f(x) = y_2$

Since f is a function,

$$y_1 = y_2$$

- **Surjectivity:** Since f is a function,

$$\begin{aligned} \forall x \in X, \exists y \in Y, f(x) = y \\ \Downarrow \\ f^{-1}(y) = x \\ \Downarrow \\ \forall x \in X, \exists y \in Y, f^{-1}(y) = x \end{aligned}$$

□

Definition 2.1.6. Let $E \subseteq X$, $F \subseteq Y$, and $f : X \rightarrow Y$ be given. The following **restriction functions** are defined as

- The **restriction** $f|_E : E \rightarrow Y$ of f to E is defined as

$$\forall x \in E, f|_E(x) = f(x)$$

- The **corestriction** $f|_F^F : X \rightarrow F$ of f to F is defined as

$$\forall x \in X, f|_F^F(x) = f(x)$$

for $f(X) \subseteq F$

- The **bi-restriction** $f|_E^F : E \rightarrow F$ of f to E and F is defined as

$$\forall x \in E, f|_E^F(x) = f(x)$$

for $f(E) \subseteq F$

Theorem 2.1.7. Let $f : X \rightarrow Y$ be given. Let $E \subseteq X$ and $F \subseteq Y$ be such that

$$f(E) \subseteq F$$

Then the following properties hold:

(A) f is injective $\implies f|_E^F$ is injective

(B) $f(E) = F \implies f|_E^F$ is surjective

(C) If

- f is injective
- $f(E) = F$

then $f|_E^F$ is bijective

Proof. Let $x_1, x_2 \in E$ be given.

(A) If f is injective, then

$$\begin{aligned} f|_E^F(x_1) &= f|_E^F(x_2) \\ &\Downarrow \\ f(x_1) &= f(x_2) \\ &\Downarrow \\ x_1 &= x_2 \end{aligned}$$

(B) If $f(E) = F$, then

$$\begin{aligned} \forall y \in F, \exists x \in E, f(x) &= y \\ &\Downarrow \\ \forall y \in F, \exists x \in E, f|_E^F(x) &= y \end{aligned}$$

(C) follows from (A) and (B)

□

Definition 2.1.8. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be given. The **composition** $g \circ f : X \rightarrow Z$ of g with f is defined as

$$\forall x \in X, (g \circ f)(x) = g(f(x))$$

Theorem 2.1.9. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be given. Then the following properties hold:

(A) f and g are injective $\implies g \circ f$ is injective

(B) f and g are surjective $\implies g \circ f$ is surjective

(C) f and g are bijective $\implies g \circ f$ is bijective

Proof. Let $x_1, x_2 \in X$ be given.

(A) If f and g are injective, then

$$\begin{aligned} (g \circ f)(x_1) &= (g \circ f)(x_2) \\ \Downarrow \\ g(f(x_1)) &= g(f(x_2)) \\ \Downarrow \\ f(x_1) &= f(x_2) \\ \Downarrow \\ x_1 &= x_2 \end{aligned}$$

(B) If f and g are surjective, then

$$\forall z \in Z, \exists y \in Y, g(y) = z$$

Also,

$$\forall y \in Y, \exists x \in X, f(x) = y$$

Therefore,

$$\forall z \in Z, \exists x \in X, (g \circ f)(x) = z$$

(C) follows from (A) and (B)

□

Theorem 2.1.10. Let S be a set of sets. Define

$$\forall X, Y \in S, X \sim Y \iff \exists f : X \xrightarrow{\sim} Y$$

Then $\sim$ is an equivalence relation on S .

Proof. Let $X, Y, Z \in S$ be given.

• **Reflexivity:**

$$\begin{aligned} \forall X \in S, \exists f : X &\xrightarrow{\sim} X, f(x) = x \\ \Downarrow \\ \forall X \in S, X &\sim X \end{aligned}$$

• **Symmetry:** If $X \sim Y$, by [Theorem 2.1.5](#),

$$\begin{aligned} \exists f : X &\xrightarrow{\sim} Y \\ \Downarrow \\ \exists f^{-1} : Y &\xrightarrow{\sim} X \end{aligned}$$

Therefore,

$$\forall X, Y \in S, X \sim Y \implies Y \sim X$$

- **Transitivity:** If $X \sim Y$ and $Y \sim Z$, by [Theorem 2.1.5](#),

$$\begin{array}{c} \exists f : X \xrightarrow{\sim} Y \wedge \exists g : Y \xrightarrow{\sim} Z \\ \downarrow \\ \exists g \circ f : X \xrightarrow{\sim} Z \end{array}$$

Therefore,

$$\forall X, Y, Z \in S, X \sim Y \wedge Y \sim Z \implies X \sim Z$$

□

2.2 Finite Sets

Recall the relation $\sim$ defined in [Theorem 2.1.10](#). A set X is said to be **finite** if

$$\exists n \in \mathbb{N}_0, \mathbb{N}^n \sim X$$

In this case, n is called the **cardinality** of X , which is denoted by

$$|X| = n$$

A set X is said to be

- **infinite** if it is not finite
- **countable** if $\mathbb{N} \sim X$
- **at most countable** if it is either finite or countable
- **uncountable** if it is neither finite nor countable

Remark. In this textbook, we will use the following notations:

- X is finite $\iff |X| < \aleph_0$
- X is infinite $\iff |X| \geq \aleph_0$
- X is countable $\iff |X| = \aleph_0$
- X is at most countable $\iff |X| \leq \aleph_0$
- X is uncountable $\iff |X| > \aleph_0$

The order symbols above are not **total orders**; they are just convenient notations.

Lemma 2.2.1.

$$\forall n \in \mathbb{N}_0, \forall A \subseteq \mathbb{N}^n, |A| < \aleph_0$$

Proof. Define

$$P(n) : \forall A \subseteq \mathbb{N}^n, |A| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}_0$.

- **Base case:** Define $f : \mathbb{N}^0 \xrightarrow{\sim} \emptyset$ as

$$\forall k \in \mathbb{N}^0, f(k) = k$$

Then

$$|\emptyset| = 0 < \aleph_0$$

Therefore, $P(0)$ is true.

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}_0$. Let $A \subseteq \mathbb{N}^{n+1}$ be given.

- If $n+1 \notin A$, then

$$\begin{array}{c} A \subseteq \mathbb{N}^n \\ \downarrow \\ |A| < \aleph_0 \end{array}$$

- If $n+1 \in A$, define

$$B := A \setminus \{n+1\} \subseteq \mathbb{N}^n$$

Since $B \subseteq \mathbb{N}^n$,

$$\begin{array}{c} B \subseteq \mathbb{N}^n \\ \downarrow \\ |B| < \aleph_0 \\ \downarrow \\ \exists m \in \mathbb{N}_0, \exists f : \mathbb{N}^m \xrightarrow{\sim} B \end{array}$$

Define $g : \mathbb{N}^{m+1} \xrightarrow{\sim} A$ as

$$\forall k \in \mathbb{N}^{m+1}, g(k) = \begin{cases} f(k) & \text{if } k \leq m \\ n+1 & \text{if } k = m+1 \end{cases}$$

Then

$$|A| = m+1 < \aleph_0$$

Therefore, $P(n+1)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}_0$. □

Theorem 2.2.2.

$$\forall X, \forall Y \subseteq X, |X| < \aleph_0 \implies |Y| < \aleph_0$$

Proof. Let X be finite. Then

$$\begin{aligned} |X| &< \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n &\xrightarrow{\sim} X \end{aligned}$$

Define

$$Z := f^{-1}(Y) \subseteq \mathbb{N}^n$$

for $Y \subseteq X$. By [Lemma 2.2.1](#),

$$\begin{aligned} Z &\subseteq \mathbb{N}^n \\ \Downarrow \\ |Z| &< \aleph_0 \\ \Downarrow \\ \exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m &\xrightarrow{\sim} Z \end{aligned}$$

By [Theorems 2.1.7](#) and [2.1.9](#),

$$\begin{aligned} f(Z) &= Y \\ \Downarrow \\ \exists f|_Z^Y : Z &\xrightarrow{\sim} Y \\ \Downarrow \\ \exists f|_Z^Y \circ g : \mathbb{N}^m &\xrightarrow{\sim} Y \\ \Downarrow \\ |Y| &< \aleph_0 \end{aligned}$$

□

Lemma 2.2.3.

$$\forall Y, (\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \rightarrow Y) \implies |Y| < \aleph_0$$

Proof. Assume that

$$\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \rightarrow Y$$

Define

$$N_y := f^{-1}(\{y\}) \subseteq \mathbb{N}^n$$

for each $y \in Y$. By [Theorem 1.1.4](#),

$$\begin{aligned} f \text{ is surjective} \\ \Downarrow \\ \forall y \in Y, N_y &\neq \emptyset \\ \Downarrow \\ \exists n_y \in N_y, \forall x \in N_y, &n_y \leq x \end{aligned}$$

Define

$$M_Y := \{n_y \mid y \in Y\} \subseteq \mathbb{N}^n$$

Define $g : M_Y \xrightarrow{\sim} Y$ as

$$\forall n_y \in M_Y, g(n_y) = y$$

By [Lemma 2.2.1](#) and [theorem 2.1.9](#),

$$\begin{aligned} |M_Y| &< \aleph_0 \\ \Downarrow \\ \exists m \in \mathbb{N}_0, \exists h : \mathbb{N}^m &\xrightarrow{\sim} M_Y \\ \Downarrow \\ \exists g \circ h : \mathbb{N}^m &\xrightarrow{\sim} Y \\ \Downarrow \\ |Y| &< \aleph_0 \end{aligned}$$

□

Theorem 2.2.4.

$$\forall X \forall Y, (|X| < \aleph_0 \wedge \exists f : X \rightarrow Y) \implies |Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge \exists f : X \rightarrow Y$$

By [Theorem 2.1.9](#) and [lemma 2.2.3](#),

$$\begin{aligned} |X| &< \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}_0, \exists g : \mathbb{N}^n &\xrightarrow{\sim} X \\ \Downarrow \\ \exists f \circ g : \mathbb{N}^n &\rightarrow Y \\ \Downarrow \\ |Y| &< \aleph_0 \end{aligned}$$

□

Theorem 2.2.5. *Let X satisfy*

- $0 < |X| < \aleph_0$
- X is ordered

Then

- $\exists m \in X, \forall x \in X, m \leq x$

- $\exists M \in X, \forall x \in X, x \leq M$

Proof. Since $0 < |X| < \aleph_0$,

$$\begin{aligned} 0 < |X| < \aleph_0 \\ \Downarrow \\ \exists n \in \mathbb{N}, \exists f : \mathbb{N}^n \xrightarrow{\sim} X \end{aligned}$$

The elements of X can be listed as

$$X = \{a_1, a_2, \dots, a_n\}$$

where

$$\forall k \in \mathbb{N}, a_k = f(k)$$

Define $P(n)$: If

- $|X| = n$
- X is ordered

then

- $\exists m \in X, \forall x \in X, m \leq x$
- $\exists M \in X, \forall x \in X, x \leq M$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $X = \{a_1\}$ be given.

- $\forall x \in X, a_1 \leq x$
- $\forall x \in X, x \leq a_1$

Therefore, $P(1)$ is true.

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Let

- $X = \{a_1, a_2, \dots, a_{n+1}\}$
- $Y = \{a_1, a_2, \dots, a_n\} \subseteq X$

be given. Then

- $\exists k \in \mathbb{N}^n, \forall y \in Y, a_k \leq y$
- $\exists K \in \mathbb{N}^n, \forall y \in Y, y \leq a_K$

Also,

- $\forall x \in X, \min\{a_k, a_{n+1}\} \leq x$
- $\forall x \in X, x \leq \max\{a_K, a_{n+1}\}$

Therefore, $P(n + 1)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Lemma 2.2.6. *Let $n, m \in \mathbb{N}_0$ be given. Then*

$$(A) (\exists f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$$

$$(B) (\exists f : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies n \geq m$$

$$(C) (\exists f : \mathbb{N}^n \xrightarrow{\sim} \mathbb{N}^m) \implies n = m$$

(C) guarantees the uniqueness of the cardinality.

$$\begin{aligned} n = |X| \wedge m = |X| \\ \Downarrow \\ \mathbb{N}^n \sim X \wedge \mathbb{N}^m \sim X \\ \Downarrow \\ \mathbb{N}^n \sim \mathbb{N}^m \\ \Downarrow \\ n = m \end{aligned}$$

Proof. (A) Define

$$P(n) : \forall m \in \mathbb{N}_0, (\exists f_n : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}_0$.

• **Base case:**

$$\begin{aligned} \forall m \in \mathbb{N}_0, 0 \leq m \\ \Downarrow \\ P(0) \text{ is true} \end{aligned}$$

• **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}_0$. If $m = 0$, then

$$\nexists f_{n+1} : \mathbb{N}^{n+1} \hookrightarrow \mathbb{N}^0$$

Otherwise, let $f_{n+1} : \mathbb{N}^{n+1} \hookrightarrow \mathbb{N}^m$ be given with

$$f_{n+1}(n + 1) = x \in \mathbb{N}^m$$

Define $g_m : \mathbb{N}^m \xrightarrow{\sim} \mathbb{N}^m$ as

$$\forall y \in \mathbb{N}^m, g_m(y) = \begin{cases} x, & \text{if } y = m \\ m, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Define $f_n : \mathbb{N}^n \rightarrow \mathbb{N}^{m-1}$ as

$$f_n \equiv (g_m \circ f_{n+1})|_{\mathbb{N}^n}^{\mathbb{N}^{m-1}}$$

By [Theorems 2.1.7](#) and [2.1.9](#),

$$\begin{aligned} f_n &\text{ is injective} \\ &\Downarrow \\ n &\leq m-1 \\ &\Downarrow \\ n+1 &\leq m \\ &\Downarrow \\ P(n+1) &\text{ is true} \end{aligned}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}_0$.

(B) Define

$$Q(m) : \forall n \in \mathbb{N}_0, (\exists f_m : \mathbb{N}^n \rightarrow \mathbb{N}^m) \implies n \geq m$$

We will show that $Q(m)$ is true for all $m \in \mathbb{N}_0$.

• **Base case:**

$$\begin{aligned} \forall n \in \mathbb{N}_0, n \geq 0 \\ &\Downarrow \\ Q(0) &\text{ is true} \end{aligned}$$

• **Inductive step:** Let $Q(m)$ be true for $m \in \mathbb{N}_0$. If $n = 0$, then

$$\nexists f_{m+1} : \mathbb{N}^0 \rightarrow \mathbb{N}^{m+1}$$

Otherwise, let $f_{m+1} : \mathbb{N}^n \rightarrow \mathbb{N}^{m+1}$ be given with

$$f_{m+1}(n) = x \in \mathbb{N}^{m+1}$$

Define $g_{m+1} : \mathbb{N}^{m+1} \xrightarrow{\sim} \mathbb{N}^{m+1}$ as

$$\forall y \in \mathbb{N}^{m+1}, g_{m+1}(y) = \begin{cases} x, & \text{if } y = m+1 \\ m+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Define

$$K_{m+1} := (g_{m+1} \circ f_{m+1})^{-1}(\{m+1\}) \subseteq \mathbb{N}^n$$

Define $f_m : \mathbb{N}^{n-1} \rightarrow \mathbb{N}^m$ as

$$\forall y \in \mathbb{N}^{n-1}, f_m(y) = \begin{cases} (g_{m+1} \circ f_{m+1})(y), & \text{if } y \in \mathbb{N}^{n-1} \setminus K_{m+1} \\ m, & \text{if } y \in K_{m+1} \end{cases}$$

By [Theorem 2.1.9](#),

$$\begin{aligned}
 &f_m \text{ is surjective} \\
 &\Downarrow \\
 &n - 1 \geq m \\
 &\Downarrow \\
 &n \geq m + 1 \\
 &\Downarrow \\
 &Q(m + 1) \text{ is true}
 \end{aligned}$$

By the *principle of mathematical induction*, $Q(m)$ is true for all $m \in \mathbb{N}_0$.

(C) follows from (A) and (B). □

Lemma 2.2.7. *Let $n, m \in \mathbb{N}_0$ be given. Then*

$$(A) \quad n \leq m \implies \exists f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m$$

$$(B) \quad n \geq m \implies \exists f : \mathbb{N}^n \rightarrow \mathbb{N}^m$$

$$(C) \quad n = m \implies \exists f : \mathbb{N}^n \xrightarrow{\sim} \mathbb{N}^m$$

This lemma is the converse of [Lemma 2.2.6](#).

Proof. (A) If $n \leq m$, define $f : \mathbb{N}^n \hookrightarrow \mathbb{N}^m$ as

$$\forall i \in \mathbb{N}^n, f(i) = i$$

(B) If $n \geq m$, define $f : \mathbb{N}^n \rightarrow \mathbb{N}^m$ as

$$\forall i \in \mathbb{N}^n, f(i) = \begin{cases} i, & \text{if } i \leq m \\ m, & \text{if } i > m \end{cases}$$

(C) follows from (A) and (B). □

Theorem 2.2.8. *Let X, Y be finite. Then*

$$(A) \quad |X| \leq |Y| \iff \exists f : X \hookrightarrow Y$$

$$(B) \quad |X| \geq |Y| \iff \exists f : X \rightarrow Y$$

$$(C) \quad |X| = |Y| \iff \exists f : X \xrightarrow{\sim} Y$$

*The contraposition of (A, $\iff$) is called the **pigeonhole principle**.*

Proof. Since X and Y are finite,

- $\exists n \in \mathbb{N}_0, \exists f_X : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists f_Y : \mathbb{N}^m \xrightarrow{\sim} Y$

($\Leftarrow$) Define $f : X \rightarrow Y$ and $F : \mathbb{N}^n \rightarrow \mathbb{N}^m$ as

$$F \equiv (f_Y)^{-1} \circ f \circ f_X$$

By [Theorems 2.1.5](#) and [2.1.9](#),

- f is injective $\implies F$ is injective
- f is surjective $\implies F$ is surjective

By [Lemma 2.2.6](#),

$$(A) (\exists f : X \hookrightarrow Y) \implies (\exists F : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies n \leq m$$

$$(B) (\exists f : X \twoheadrightarrow Y) \implies (\exists F : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies n \geq m$$

(C) follows from (A) and (B).

($\implies$) Define $f : X \rightarrow Y$ and $G : \mathbb{N}^m \rightarrow \mathbb{N}^n$ as

$$f \equiv f_Y \circ G \circ (f_X)^{-1}$$

By [Theorems 2.1.5](#) and [2.1.9](#),

- (A) G is injective $\implies f$ is injective
- (B) G is surjective $\implies f$ is surjective

By [Lemma 2.2.7](#),

$$(A) n \leq m \implies (\exists G : \mathbb{N}^n \hookrightarrow \mathbb{N}^m) \implies (\exists f : X \hookrightarrow Y)$$

$$(B) n \geq m \implies (\exists G : \mathbb{N}^n \twoheadrightarrow \mathbb{N}^m) \implies (\exists f : X \twoheadrightarrow Y)$$

(C) follows from (A) and (B).

□

Lemma 2.2.9.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \wedge X \cap Y = \emptyset \implies |X \cup Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0 \wedge X \cap Y = \emptyset$$

Then

- $\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m \xrightarrow{\sim} Y$

Define $h : \mathbb{N}^{m+n} \xrightarrow{\sim} X \cup Y$ as

$$\forall i \in \mathbb{N}^{m+n}, h(i) = \begin{cases} f(i), & \text{if } i \leq n \\ g(i - n), & \text{if } n < i \leq n + m \end{cases}$$

Then

$$|X \cup Y| = m + n < \aleph_0$$

□

Theorem 2.2.10.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0$$

Define

- $A = X \setminus Y \subseteq X$
- $B = Y \setminus X \subseteq Y$
- $C = X \cap Y \subseteq X$

By [Theorem 2.2.2](#),

- $|A| < \aleph_0, |B| < \aleph_0$, and $|C| < \aleph_0$
- $A \cap B = \emptyset, B \cap C = \emptyset$, and $C \cap A = \emptyset$

Consider the following:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$

By [Lemma 2.2.9](#),

$$\begin{aligned} |A \cup B| &< \aleph_0 \\ \downarrow \\ |(A \cup B) \cup C| &< \aleph_0 \\ \downarrow \\ |X \cup Y| &< \aleph_0 \end{aligned}$$

□

Theorem 2.2.11.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left| \bigcup_{k=1}^n X_k \right| < \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left| \bigcup_{k=1}^n X_k \right| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| < \aleph_0$ be given. Then

$$\begin{array}{c} |X_1| < \aleph_0 \\ \Downarrow \\ \left| \bigcup_{k=1}^1 X_k \right| < \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{array}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| < \aleph_0$$

By the inductive hypothesis,

$$\begin{array}{l} - \left| \bigcup_{k=1}^n X_k \right| < \aleph_0 \\ - |X_{n+1}| < \aleph_0 \end{array}$$

By [Theorem 2.2.10](#),

$$\begin{array}{c} \bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^n X_k \right) \cup X_{n+1} \\ \Downarrow \\ \left| \bigcup_{k=1}^{n+1} X_k \right| < \aleph_0 \\ \Downarrow \\ P(n+1) \text{ is true} \end{array}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.2.12.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \times Y| < \aleph_0$$

Proof. Assume that

$$|X| < \aleph_0 \wedge |Y| < \aleph_0$$

Then

- $\exists n \in \mathbb{N}_0, \exists f : \mathbb{N}^n \xrightarrow{\sim} X$
- $\exists m \in \mathbb{N}_0, \exists g : \mathbb{N}^m \xrightarrow{\sim} Y$

Define

$$i = (a - 1)m + b \in \mathbb{N}^{nm}$$

for each $a \in \mathbb{N}^n$ and $b \in \mathbb{N}^m$. Define $h : \mathbb{N}^{nm} \xrightarrow{\sim} X \times Y$ as

$$\forall i \in \mathbb{N}^{nm}, h(i) = (f(a), g(b))$$

Then

$$|X \times Y| = nm < \aleph_0$$

□

Theorem 2.2.13.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left| \prod_{k=1}^n X_k \right| < \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| < \aleph_0) \implies \left| \prod_{k=1}^n X_k \right| < \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| < \aleph_0$ be given. Then

$$\begin{array}{c} |X_1| < \aleph_0 \\ \downarrow \\ \left| \prod_{k=1}^1 X_k \right| < \aleph_0 \\ \downarrow \\ P(1) \text{ is true} \end{array}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| < \aleph_0$$

By the inductive hypothesis,

$$\begin{aligned} - \left| \prod_{k=1}^n X_k \right| &< \aleph_0 \\ - |X_{n+1}| &< \aleph_0 \end{aligned}$$

By [Lemma 2.2.12](#),

$$\begin{aligned} \prod_{k=1}^{n+1} X_k &= \left(\prod_{k=1}^n X_k \right) \times X_{n+1} \\ &\Downarrow \\ \left| \prod_{k=1}^{n+1} X_k \right| &< \aleph_0 \\ &\Downarrow \\ P(n+1) &\text{ is true} \end{aligned}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.2.14.

$$\forall X, \nexists f : X \rightarrow \mathcal{P}(X)$$

where

$$\mathcal{P}(X) := \{E \mid E \subseteq X\}$$

is called the **power set** of X . This theorem is called **Cantor's theorem**.

Proof. Assume that

$$\exists f : X \rightarrow \mathcal{P}(X)$$

Define

$$S := \{x \in X \mid x \notin f(x)\} \subseteq X$$

Then

$$\begin{aligned} S &\subseteq X \\ &\Downarrow \\ S &\in \mathcal{P}(X) \\ &\Downarrow \\ \exists x \in X, f(x) &= S \end{aligned}$$

There are two cases:

- If $x \in S$, then

$$\begin{array}{c}
 x \in S \\
 \Downarrow \\
 x \notin f(x) = S \\
 \Downarrow \\
 \perp
 \end{array}$$

- If $x \notin S$, then

$$\begin{array}{c}
 x \notin S \\
 \Downarrow \\
 x \in f(x) = S \\
 \Downarrow \\
 \perp
 \end{array}$$

Therefore,

$$\nexists f : X \rightarrow \mathcal{P}(X)$$

□

2.3 Countable and Uncountable Sets

Theorem 2.3.1.

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

Proof. Assume that

$$\exists X \exists Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \wedge |X \setminus Y| < \aleph_0$$

By [Theorem 2.2.10](#),

$$|(X \setminus Y) \cup Y| < \aleph_0$$

By [Theorem 2.2.2](#),

$$\begin{array}{c}
 X = (X \setminus Y) \cup (X \cap Y) \\
 \Downarrow \\
 X \subseteq (X \setminus Y) \cup Y \\
 \Downarrow \\
 |X| < \aleph_0 \\
 \Downarrow \\
 \perp
 \end{array}$$

Therefore,

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

□

Theorem 2.3.2.

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge (\exists f : X \hookrightarrow Y) \implies |Y| \geq \aleph_0$$

Proof. Assume that

$$\exists X \exists Y, |X| \geq \aleph_0 \wedge (\exists f : X \hookrightarrow Y) \wedge |Y| < \aleph_0$$

Define $g : Y \rightarrow X$ as

$$\forall y \in Y, g(y) = \begin{cases} f^{-1}(y) & \text{if } y \in f(X) \\ f^{-1}(y_0) & \text{otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. By [Theorem 2.2.4](#),

$$\begin{array}{c} \exists g : Y \rightarrow X \\ \Downarrow \\ |X| < \aleph_0 \\ \Downarrow \\ \perp \end{array}$$

Therefore,

$$\forall X \forall Y, |X| \geq \aleph_0 \wedge (\exists f : X \hookrightarrow Y) \implies |Y| \geq \aleph_0$$

□

Lemma 2.3.3.

$$|\mathbb{N}| \geq \aleph_0$$

Proof. Assume that

$$|\mathbb{N}| < \aleph_0$$

By [Theorem 2.2.5](#),

$$\exists m \in \mathbb{N}, \forall x \in \mathbb{N}, x \leq m$$

Then

$$\begin{array}{c} m < m + 1 \\ \Downarrow \\ m + 1 \notin \mathbb{N} \\ \Downarrow \\ \perp \end{array}$$

Therefore,

$$|\mathbb{N}| \geq \aleph_0$$

□

Theorem 2.3.4.

$$\forall X, |X| = \aleph_0 \implies |X| \geq \aleph_0$$

Proof. Let $|X| = \aleph_0$ be given. Then

$$\exists f : \mathbb{N} \xrightarrow{\sim} X$$

By [Theorem 2.3.2](#) and [lemma 2.3.3](#),

$$|X| \geq \aleph_0$$

□

Theorem 2.3.5.

$$|\mathbb{Z}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \rightarrow \mathbb{Z}$ as

$$\forall n \in \mathbb{N}, f(n) = \begin{cases} -\frac{n}{2} & \text{if } n \text{ is even} \\ \frac{n-1}{2} & \text{if } n \text{ is odd} \end{cases}$$

We will show that

$$\begin{array}{c} f \text{ is bijective} \\ \Downarrow \\ |\mathbb{Z}| = \aleph_0 \end{array}$$

• **Injectivity:** Let $f(n_1) = f(n_2)$ be given for $n_1, n_2 \in \mathbb{N}$. Since

- $n \text{ is even} \implies f(n) = -\frac{n}{2} < 0$
- $n \text{ is odd} \implies f(n) = \frac{n-1}{2} \geq 0$

n_1 and n_2 are both even or both odd.

- If both n_1 and n_2 are even, then

$$\begin{array}{c} -\frac{n_1}{2} = -\frac{n_2}{2} \\ \Downarrow \\ n_1 = n_2 \end{array}$$

- If both n_1 and n_2 are odd, then

$$\begin{array}{c} \frac{n_1-1}{2} = \frac{n_2-1}{2} \\ \Downarrow \\ n_1 = n_2 \end{array}$$

• **Surjectivity:** Let $k \in \mathbb{Z}$ be given.

– If $k \geq 0$, then

$$\begin{aligned} 2k + 1 &\in \mathbb{N} \\ \Downarrow \\ f(2k + 1) &= \frac{(2k + 1) - 1}{2} = \frac{2k}{2} = k \end{aligned}$$

– If $k < 0$, then

$$\begin{aligned} -2k &\in \mathbb{N} \\ \Downarrow \\ f(-2k) &= -\frac{-2k}{2} = k \end{aligned}$$

□

Lemma 2.3.6.

$$\forall X \subseteq \mathbb{N}, |X| \geq \aleph_0 \implies |X| = \aleph_0$$

Proof. Assume that

- $X \subseteq \mathbb{N}$
- $|X| \geq \aleph_0$

Define $(X_n)_{n=1}^\infty$ and $f : \mathbb{N} \rightarrow X$ as

$$\begin{array}{ll} X_1 = X \subseteq \mathbb{N} & f(1) = \min X_1 \\ X_2 = X_1 \setminus \{f(1)\} \subseteq \mathbb{N} & f(2) = \min X_2 \\ \vdots & \vdots \\ X_{n+1} = X_n \setminus \{f(n)\} \subseteq \mathbb{N} & f(n+1) = \min X_{n+1} \end{array}$$

for each $n \in \mathbb{N}$. By [Theorems 1.1.4](#) and [2.3.1](#),

$$\begin{aligned} \forall n \in \mathbb{N}, |X_n| &\geq \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, X_n &\neq \emptyset \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f(n) &= \min X_n \in X_n \\ \Downarrow \\ f &\text{ is well-defined} \end{aligned}$$

We will show that

$$\begin{aligned} f &\text{ is bijective} \\ \Downarrow \\ |X| &= \aleph_0 \end{aligned}$$

- **Injectivity:** Let $n_1 \neq n_2$ (WLOG $n_1 < n_2$) be given for $n_1, n_2 \in \mathbb{N}$.

- $f(n_1) = \min X_{n_1} \in X_{n_1}$
- $f(n_2) = \min X_{n_2} \in X_{n_2}$

Then

$$\begin{aligned} X_{n_2} &\subseteq X_{n_1} \setminus \{f(n_1)\} \\ &\Downarrow \\ f(n_2) &\neq f(n_1) \\ &\Downarrow \\ f &\text{ is injective} \end{aligned}$$

- **Surjectivity:** Let $x \in X$ be given. Define

$$S := \{n \in X \mid n < x\}$$

By [Lemma 2.2.1](#),

$$\begin{aligned} S &\subseteq \mathbb{N}^{x-1} \\ &\Downarrow \\ |S| &< \aleph_0 \\ &\Downarrow \\ \exists k \in \mathbb{N}_0, |S| &= k \end{aligned}$$

If $k = 0$, then

- $S = \emptyset$
- $x = \min X_1 = f(1)$

Otherwise,

- $S = \{s_1, s_2, \dots, s_k\}$
- $s_1 < s_2 < \dots < s_k$

Then

$$\begin{array}{ll} \min X_1 = \min S & X_2 = X \setminus \{s_1\} \\ \min X_2 = \min S \setminus \{s_1\} & X_3 = X \setminus \{s_1, s_2\} \\ \vdots & \vdots \\ \min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} & X_{k+1} = X \setminus S \end{array}$$

Therefore,

$$\begin{aligned} f(k+1) &= \min X_{k+1} = x \\ &\Downarrow \\ f &\text{ is surjective} \end{aligned}$$

□

Theorem 2.3.7.

$$\forall X, \forall Y \subseteq X, |X| = \aleph_0 \wedge |Y| \geq \aleph_0 \implies |Y| = \aleph_0$$

Proof. Assume that

- $Y \subseteq X$
- $|X| = \aleph_0$
- $|Y| \geq \aleph_0$

By Theorem 2.1.7,

$$\begin{array}{c} \exists f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \\ \exists f|_{f^{-1}(Y)}^Y : f^{-1}(Y) \xrightarrow{\sim} Y \end{array}$$

By Theorem 2.2.4, if $f^{-1}(Y)$ is finite, then

$$\begin{array}{c} |f^{-1}(Y)| < \aleph_0 \\ \Downarrow \\ |Y| < \aleph_0 \\ \Downarrow \\ \perp \end{array}$$

Therefore,

$$|f^{-1}(Y)| \geq \aleph_0$$

By Lemma 2.3.6,

$$\begin{array}{c} |f^{-1}(Y)| \subseteq \mathbb{N} \\ \Downarrow \\ |f^{-1}(Y)| = \aleph_0 \\ \Downarrow \\ \exists g : \mathbb{N} \xrightarrow{\sim} f^{-1}(Y) \\ \Downarrow \\ \exists f|_{f^{-1}(Y)}^Y \circ g : \mathbb{N} \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| = \aleph_0 \end{array}$$

□

Theorem 2.3.8.

$$\forall X, |X| \geq \aleph_0 \wedge (\exists f : \mathbb{N} \twoheadrightarrow X) \implies |X| = \aleph_0$$

Proof. Assume that

- $|X| \geq \aleph_0$
- $\exists f : \mathbb{N} \rightarrow X$

Define

$$S_k := f^{-1}(\{k\})$$

for each $k \in X$. By [Theorem 1.1.4](#),

$$\begin{aligned} f \text{ is surjective} \\ \Downarrow \\ \forall k \in X, S_k \neq \emptyset \\ \Downarrow \\ \forall k \in X, \exists m \in S_k, m = \min S_k \end{aligned}$$

Define $g : X \rightarrow \mathbb{N}$ as

$$\forall k \in X, g(k) = \min S_k$$

Let $k_1, k_2 \in X$ be given. Then

$$\begin{aligned} \forall i, j \in X, i \neq j &\implies S_i \cap S_j = \emptyset \\ \Downarrow \\ g(k_1) = g(k_2) &\implies \min S_{k_1} = \min S_{k_2} \implies k_1 = k_2 \\ \Downarrow \\ g \text{ is injective} \end{aligned}$$

By [Theorem 2.1.7](#),

$$\exists g|^{g(X)} : X \xrightarrow{\sim} g(X)$$

By [Theorem 2.3.2](#),

$$\begin{aligned} |X| &\geq \aleph_0 \\ \Downarrow \\ |g(X)| &\geq \aleph_0 \end{aligned}$$

By [Lemma 2.3.6](#),

$$\begin{aligned} g(X) &\subseteq \mathbb{N} \\ \Downarrow \\ |g(X)| &= \aleph_0 \\ \Downarrow \\ |X| &= \aleph_0 \end{aligned}$$

□

Theorem 2.3.9.

$$(\forall n \in \mathbb{N}, 0 < |X_n| < \aleph_0) \implies \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0$$

Proof. Assume that

$$\begin{aligned} \forall n \in \mathbb{N}, 0 < |X_n| < \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists m_n \in \mathbb{N}, \exists g_n : \mathbb{N}^{m_n} \xrightarrow{\sim} X_n \end{aligned}$$

Define $h : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$ as

- $\forall m \in \mathbb{N}_{m_1}, h(m) = g_1(m)$
- $\forall n \in \mathbb{N}_2, \forall m \in \mathbb{N}_{m_n}, h\left(\sum_{k=1}^{n-1} m_k + m\right) = g_n(m)$

By [Theorem 2.3.8](#),

$$\left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0$$

□

Theorem 2.3.10.

$$(\forall n \in \mathbb{N}, |X_n| = \aleph_0) \implies \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0$$

Proof. Define

$$T_n := \left\{ k \in \mathbb{N} \mid n \leq \frac{k(k+1)}{2} \right\}$$

for each $n \in \mathbb{N}$. By [Theorem 1.1.4](#),

$$\begin{aligned} 2n &\in T_n \\ \Downarrow \\ T_n &\neq \emptyset \\ \Downarrow \\ \exists t_n &\in T_n, t_n = \min T_n \end{aligned}$$

for all $n \in \mathbb{N}$. Define

$$s_n := n - \frac{t_n(t_n - 1)}{2} = n - \frac{t_n(t_n + 1)}{2} + t_n \in \mathbb{N}$$

for each $n \in \mathbb{N}$. Then

$$\begin{aligned} \frac{t_n(t_n - 1)}{2} &< n \leq \frac{t_n(t_n + 1)}{2} \\ \Downarrow \\ t_n - s_n + 1 &= \frac{t_n(t_n + 1)}{2} - n + 1 \geq 1 \\ \Downarrow \\ t_n - s_n + 1 &\in \mathbb{N} \end{aligned}$$

for all $n \in \mathbb{N}$. Assume that

$$\begin{aligned} \forall n \in \mathbb{N}, |X_n| &= \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f_n : \mathbb{N} &\xrightarrow{\sim} X_n \end{aligned}$$

Define $g : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$ as

$$\forall n \in \mathbb{N}, g(n) = f_{s_n}(t_n - s_n + 1)$$

Let $x \in \bigcup_{n=1}^{\infty} X_n$ be given. Then

$$\begin{aligned} \exists m \in \mathbb{N}, x &\in X_m \\ \Downarrow \\ \exists l \in \mathbb{N}, f_m(l) &= x \end{aligned}$$

Define

$$k := \frac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}$$

We will show that

$$\begin{aligned} l+m-2 &\notin T_k \wedge l+m-1 \in T_k \\ \Downarrow \\ t_k &= \min T_k = l+m-1 \\ \Downarrow \\ s_k &= k - \frac{t_k(t_k-1)}{2} = m \\ \Downarrow \\ g(k) &= f_{s_k}(t_k - s_k + 1) = f_m(l+m-1-m+1) = f_m(l) = x \end{aligned}$$

Consider the following inequalities:

- Let $z = l+m-2$ be given. Then

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0 \end{aligned}$$

Therefore,

$$z = l+m-2 \notin T_k$$

- Let $z = l + m - 1$ be given. Then

$$\begin{aligned}
 k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\
 &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\
 &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\
 &= m - (l+m-1) = 1 - l \leq 0
 \end{aligned}$$

Therefore,

$$z = l + m - 1 \in T_k$$

By [Theorem 2.3.8](#),

g is surjective

$$\begin{aligned}
 &\Downarrow \\
 &\left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0
 \end{aligned}$$

□

Corollary 2.3.11.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left| \bigcup_{k=1}^n X_k \right| = \aleph_0$$

Proof. Assume that

$$(\forall k \in \mathbb{N}^n, |X_k| = \aleph_0)$$

for $n \in \mathbb{N}$. Define

$$X_k := X_n$$

for all $k \in \mathbb{N}_{n+1}$. By [Theorem 2.3.10](#),

$$\begin{aligned}
 &\forall n \in \mathbb{N}, |X_n| = \aleph_0 \\
 &\Downarrow \\
 &\left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \bigcup_{k=1}^n X_k &= \bigcup_{n=1}^{\infty} X_n \\
 &\Downarrow \\
 \left| \bigcup_{k=1}^n X_k \right| &= \aleph_0
 \end{aligned}$$

□

Lemma 2.3.12.

$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ as

$$\forall m, n \in \mathbb{N}, f(m, n) = \frac{(m+n-1)(m+n-2)}{2} + m$$

We will show that f is bijective.

- **Injectivity:** Let $(m_1, n_1) \neq (m_2, n_2)$ be given for $m_1, n_1, m_2, n_2 \in \mathbb{N}$. If

$$m_1 + n_1 = m_2 + n_2$$

then

$$\begin{aligned} m_1 &\neq m_2 \\ \Downarrow \\ f(m_1, n_1) &\neq f(m_2, n_2) \end{aligned}$$

Otherwise, $m_1 + n_1 \neq m_2 + n_2$ (*WLOG* $m_1 + n_1 < m_2 + n_2$). Then

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\ &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\ &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\ &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\ &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2) \end{aligned}$$

- **Surjectivity:** Let $k \in \mathbb{N}$ be given. Define t_k, s_k as in the proof of [Theorem 2.3.10](#). Then

$$f(s_k, t_k - s_k + 1) = k$$

Therefore,

$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$

□

Theorem 2.3.13.

$$\forall X \forall Y, |X| = \aleph_0 \wedge |Y| = \aleph_0 \implies |X \times Y| = \aleph_0$$

Proof. Since X and Y are countable,

- $\exists f : \mathbb{N} \xrightarrow{\sim} X$

- $\exists g : \mathbb{N} \xrightarrow{\sim} Y$

Define $h : \mathbb{N} \times \mathbb{N} \xrightarrow{\sim} X \times Y$ as

$$\forall m, n \in \mathbb{N}, h(m, n) = (f(m), g(n))$$

By [Lemma 2.3.12](#),

$$|X \times Y| = \aleph_0$$

□

Theorem 2.3.14.

$$\forall n \in \mathbb{N}, (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left| \prod_{k=1}^n X_k \right| = \aleph_0$$

Proof. Define

$$P(n) : (\forall k \in \mathbb{N}^n, |X_k| = \aleph_0) \implies \left| \prod_{k=1}^n X_k \right| = \aleph_0$$

We will show that $P(n)$ is true for all $n \in \mathbb{N}$.

- **Base case:** Let $|X_1| = \aleph_0$ be given. Then

$$\begin{array}{c} |X_1| = \aleph_0 \\ \Downarrow \\ \left| \prod_{k=1}^1 X_k \right| = \aleph_0 \\ \Downarrow \\ P(1) \text{ is true} \end{array}$$

- **Inductive step:** Let $P(n)$ be true for $n \in \mathbb{N}$. Assume that

$$\forall k \in \mathbb{N}^{n+1}, |X_k| = \aleph_0$$

By the inductive hypothesis,

$$\begin{array}{l} - \left| \prod_{k=1}^n X_k \right| = \aleph_0 \\ - |X_{n+1}| = \aleph_0 \end{array}$$

By [Theorem 2.3.13](#),

$$\begin{aligned} \prod_{k=1}^{n+1} X_k &= \left(\prod_{k=1}^n X_k \right) \times X_{n+1} \\ &\downarrow \\ \left| \prod_{k=1}^{n+1} X_k \right| &= \aleph_0 \\ &\downarrow \\ P(n+1) &\text{ is true} \end{aligned}$$

By the *principle of mathematical induction*, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.3.15.

$$|\mathbb{Q}| = \aleph_0$$

Proof. Let $q \in \mathbb{Q}$ be given. Then

$$\exists m \in \mathbb{Z}, \exists n \in \mathbb{N}, q = \frac{m}{n}$$

Define $f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$ as

$$\forall m \in \mathbb{Z} \forall n \in \mathbb{N}, f(m, n) = \frac{m}{n}$$

By [Theorems 2.3.5](#) and [2.3.13](#),

$$\begin{aligned} |\mathbb{Z} \times \mathbb{N}| &= \aleph_0 \\ &\downarrow \\ \exists g : \mathbb{N} &\xrightarrow{\sim} \mathbb{Z} \times \mathbb{N} \end{aligned}$$

By [Lemma 2.3.3](#) and [theorem 2.2.2](#),

$$\begin{aligned} \mathbb{N} &\subseteq \mathbb{Q} \\ &\downarrow \\ |\mathbb{Q}| &\geq \aleph_0 \end{aligned}$$

By [Theorems 2.1.9](#) and [2.3.8](#),

$$\begin{aligned} \exists f \circ g : \mathbb{N} &\rightarrow \mathbb{Q} \\ &\downarrow \\ |\mathbb{Q}| &= \aleph_0 \end{aligned}$$

□

Lemma 2.3.16.

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| > \aleph_0$$

Proof. Define $(e_n)_{n=1}^{\infty}$ in $\prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall m \in \mathbb{N}, e_{nm} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{otherwise} \end{cases}$$

for each $n \in \mathbb{N}$. Define $f : \mathbb{N} \hookrightarrow \prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall n \in \mathbb{N}, f(n) = e_n$$

By [Lemma 2.3.3](#) and [theorem 2.3.2](#),

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| \geq \aleph_0$$

If $\prod_{n=1}^{\infty} \{0, 1\}$ is countable, then

$$\exists g : \mathbb{N} \xrightarrow{\sim} \prod_{n=1}^{\infty} \{0, 1\}$$

Let

- $\prod_{n=1}^{\infty} \{0, 1\} = \{a_1, a_2, a_3, \dots\}$
- $a_1 = (a_{11}, a_{12}, a_{13}, \dots)$
- $a_2 = (a_{21}, a_{22}, a_{23}, \dots)$
- $\vdots$

be given, where

$$\forall n \in \mathbb{N}, a_n = g(n)$$

Define $b = (b_1, b_2, b_3, \dots) \in \prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall n \in \mathbb{N}, b_n = 1 - a_{nn}$$

Then

$$\begin{aligned}
 & \forall n \in \mathbb{N}, b \neq a_n \\
 & \Downarrow \\
 & b \notin \{a_1, a_2, a_3, \dots\} = \prod_{n=1}^{\infty} \{0, 1\} \\
 & \Downarrow \\
 & g \text{ is not surjective} \\
 & \Downarrow \\
 & \perp
 \end{aligned}$$

Therefore,

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| > \aleph_0$$

□

Theorem 2.3.17.

$$|\mathbb{R}| > \aleph_0$$

Proof. By [Lemma 2.3.3](#) and [theorem 2.2.2](#),

$$\begin{aligned}
 & \mathbb{N} \subseteq \mathbb{R} \\
 & \Downarrow \\
 & |\mathbb{R}| \geq \aleph_0
 \end{aligned}$$

Define $f : \mathbb{N} \hookrightarrow [0, 1)$ as

$$\forall n \in \mathbb{N}, f(n) = \frac{1}{n+1}$$

By [Lemma 2.3.3](#) and [theorem 2.3.2](#),

$$|[0, 1)| \geq \aleph_0$$

By [Theorem 2.3.7](#),

$$\begin{aligned}
 & [0, 1) \subseteq \mathbb{R} \\
 & \Downarrow \\
 & |[0, 1)| > \aleph_0 \implies |\mathbb{R}| > \aleph_0
 \end{aligned}$$

Define $g : [0, 1) \rightarrow \prod_{n=1}^{\infty} \{0, 1\}$ as

$$\forall x \in [0, 1), g(x) = (a_n^x)_{n=1}^{\infty}$$

where

$(a_n)_{n=1}^{\infty}$ is the *greedy base 2 expansion* of x

Define

$$Y := g([0, 1)) \subseteq \prod_{n=1}^{\infty} \{0, 1\}$$

By Theorems 1.6.9 and 2.1.7,

$$\exists g|_Y : [0, 1) \xrightarrow{\sim} Y$$

If $[0, 1)$ is countable, then

$$\exists h : \mathbb{N} \xrightarrow{\sim} [0, 1)$$

By Theorem 2.1.9,

$$\begin{aligned} g|_Y \circ h : \mathbb{N} &\xrightarrow{\sim} Y \\ \downarrow & \\ |Y| &= \aleph_0 \end{aligned}$$

Define

$$Z := \prod_{n=1}^{\infty} \{0, 1\} \setminus Y$$

By Theorems 1.6.10 and 1.6.11,

$$Z = \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid \exists N \in \mathbb{N}, \forall n \in \mathbb{N}_N, a_n = 1 \right\}$$

Define

$$Z_N := \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid \forall n \in \mathbb{N}_N, a_n = 1 \right\}$$

for each $N \in \mathbb{N}$. Define $F_N : \prod_{n=1}^{N-1} \{0, 1\} \rightarrow Z_N$ as

$$\forall (a_n)_{n=1}^{N-1} \in \prod_{n=1}^{N-1} \{0, 1\}, F_N((a_n)_{n=1}^{N-1}) = (a_n)_{n=1}^{N-1} + (1)_{n=N}^{\infty}$$

By Theorems 2.2.4 and 2.2.13,

$$\forall N \in \mathbb{N}, 0 < |Z_N| < \aleph_0$$

By Theorem 2.3.9,

$$\begin{aligned} Z &= \bigcup_{N=1}^{\infty} Z_N \\ \downarrow & \\ |Z| &= \aleph_0 \end{aligned}$$

By [Corollary 2.3.11](#) and [lemma 2.3.16](#),

$$\begin{array}{c} \prod_{n=1}^{\infty} \{0, 1\} = Y \cup Z \\ \Downarrow \\ \left| \prod_{n=1}^{\infty} \{0, 1\} \right| = \aleph_0 \\ \Downarrow \\ \perp \end{array}$$

Therefore,

$$\begin{array}{c} |[0, 1]| > \aleph_0 \\ \Downarrow \\ |\mathbb{R}| > \aleph_0 \end{array}$$

□

2.4 Metric Spaces

A function $d : X \times X \rightarrow \mathbb{R}$ is called a **metric** on X if for all $p, q, r \in X$,

(A) $d(p, q) \geq 0$

(B) $d(p, q) = 0 \iff p = q$

(C) $d(p, q) = d(q, p)$

(D) $d(p, r) \leq d(p, q) + d(q, r)$

(D) is called the **triangle inequality**. The pair (X, d) is called a **metric space**.

Theorem 2.4.1. Let $n \in \mathbb{N}$ be given. Define $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ as

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then d is a metric on $\mathbb{R}^n$. In particular, d is called the **standard metric** on $\mathbb{R}^n$.

Proof. Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ be given.

- $d(\mathbf{x}, \mathbf{y}) \geq 0$ is trivial.
- By [Definition 1.9.1](#),

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) = 0 &\iff \|\mathbf{x} - \mathbf{y}\| = 0 \\ &\iff \forall k \in \mathbb{N}^n, x_k = y_k \\ &\iff \mathbf{x} = \mathbf{y} \end{aligned}$$

- $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ is trivial.
- By [Theorem 1.9.2](#),

$$\begin{aligned} d(\mathbf{x}, \mathbf{z}) &= \|\mathbf{x} - \mathbf{z}\| = \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\| \\ &\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\| = d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) \end{aligned}$$

Therefore, d is a metric on $\mathbb{R}^n$. □

Definition 2.4.2. Let $k \in \mathbb{N}$ be given. A set $E \subseteq \mathbb{R}^k$ is called a **k -cell** if

$$E = \left\{ \mathbf{x} \in \mathbb{R}^k \mid \forall j \in \mathbb{N}^k, a_j \leq x_j \leq b_j \right\}$$

where $-\infty < a_j \leq b_j < +\infty$ for all $j \in \mathbb{N}^k$.

Definition 2.4.3. Let (X, d) be a metric space. Let $p \in X$ and $r \in \mathbb{R}^+$ be given. The **neighborhood** of p with radius r is defined as

$$N_r(p) := \{q \in X \mid d(p, q) < r\}$$

The **deleted neighborhood** of p with radius r is defined as

$$\tilde{N}_r(p) := N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p, q) < r\}$$

In particular, if

- $X = \mathbb{R}^n$
- d is the standard metric on $\mathbb{R}^n$

then $N_r(p)$ is called the **open ball** of radius r centered at p , which is denoted by $B_r(p)$.

Definition 2.4.4. Let (X, d) be a metric space and $E \subseteq X$. Let $p \in X$ be given.

- p is called an **limit point** of E if

$$\forall r \in \mathbb{R}^+, E \cap \tilde{N}_r(p) \neq \emptyset$$

- p is called a **isolated point** of E if

$$\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}$$

- p is called an **interior point** of E if

$$\exists r \in \mathbb{R}^+, N_r(p) \subseteq E$$

Definition 2.4.5. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **open** if every point of E is an interior point of E .
- E is said to be **closed** if every limit point of E belongs to E .

Definition 2.4.6. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **bounded** if

$$\exists p \in X, \exists r \in \mathbb{R}^+, E \subseteq N_r(p)$$

- E is said to be **perfect** if every point of E is a limit point of E .
- E is said to be **dense** if every point of X is either a limit point of E or a point of E .

Theorem 2.4.7. Let (X, d) be a metric space. For every $p \in X$ and $r \in \mathbb{R}^+$, the neighborhood $N_r(p)$ is open.

Proof. Let $q \in N_r(p)$ be given. Then we have $d(p, q) < r \implies \varepsilon := r - d(p, q) > 0$. Let $s \in N_\varepsilon(q)$ be given. Then we have

$$d(q, s) < \varepsilon \implies d(p, s) \leq d(p, q) + d(q, s) < d(p, q) + \varepsilon = r.$$

Thus $s \in N_r(p)$, which implies that $N_\varepsilon(q) \subseteq N_r(p)$. Since q is arbitrary, $N_r(p)$ is open. $\square$

Theorem 2.4.8. Let (X, d) be a metric space and $E \subseteq X$. Let p be a limit point of E . Then $E \cap N_r(p)$ is infinite for all $r \in \mathbb{R}^+$.

Proof. Assume that there exists some $r \in \mathbb{R}^+$ such that $E \cap N_r(p)$ is finite. Let S be a set defined as

$$S := (E \cap N_r(p)) \setminus \{p\} = E \cap \tilde{N}_r(p).$$

Since p is a limit point of E , we have $S \neq \emptyset$. Since $E \cap N_r(p)$ is finite, $S \subseteq E \cap N_r(p)$ is also finite by [Theorem 2.2.2](#). Let T be a set defined as

$$T := \{d(p, q) \mid q \in S\} \neq \emptyset.$$

Since $p \notin S$, we have $0 \notin T$. Let $f : S \rightarrow T$ be a function defined as

$$f(q) = d(p, q).$$

Then f is surjective. By [Theorem 2.2.4](#), T is finite. Then, by [Theorem 2.2.5](#), T has a minimum element, say $\varepsilon = \min T > 0$. Then we have

- $N_\varepsilon(p) \subset N_r(p)$.
- $E \cap \tilde{N}_\varepsilon(p) = \emptyset$.

This contradicts the assumption that p is a limit point of E . Therefore, $E \cap N_r(p)$ is infinite. $\square$

Theorem 2.4.9. *Let (X, d) be a metric space and $E \subseteq X$. Then we have*

$$E \text{ is open} \iff E^c \text{ is closed.}$$

Proof. ($\implies$) Assume that E is open. Let p be a limit point of E^c . Then, for every $r \in \mathbb{R}^+$, we have

$$E^c \cap \tilde{N}_r(p) \neq \emptyset.$$

If $p \in E$, then there exists some $\varepsilon \in \mathbb{R}^+$ such that $N_\varepsilon(p) \subseteq E$. Then we have

$$E^c \cap \tilde{N}_\varepsilon(p) = \emptyset,$$

which is a contradiction. Therefore, we have $p \in E^c$, which implies that E^c is closed.

($\impliedby$) Assume that E^c is closed. Let p be a point of E . If p is not an interior point of E , then for every $r \in \mathbb{R}^+$, we have

$$E^c \cap N_r(p) \neq \emptyset.$$

Since $p \in E \implies p \notin E^c$, we have

$$E^c \cap \tilde{N}_r(p) \neq \emptyset,$$

which implies that p is a limit point of E^c . Since E^c is closed, we have $p \in E^c$, which is a contradiction. Therefore, p is an interior point of E . Since p is arbitrary, E is open. $\square$

Corollary 2.4.10. *Let (X, d) be a metric space and $E \subseteq X$. Then we have*

$$E \text{ is closed} \iff E^c \text{ is open.}$$

Theorem 2.4.11. *Let (X, d) be a metric space. Let $E_y \subseteq X$ be an open set for each $y \in Y$. Then we have*

$$\bigcup_{y \in Y} E_y \text{ is open}$$

whether Y is finite, countable, or uncountable. This theorem states that an arbitrary union of open sets is open.

Proof. Let $p \in \bigcup_{y \in Y} E_y$ be given. Then there exists some $y_0 \in Y$ such that $p \in E_{y_0}$.

Since E_{y_0} is open, there exists some $r \in \mathbb{R}^+$ such that $N_r(p) \subseteq E_{y_0}$. Then we have

$$N_r(p) \subseteq E_{y_0} \subseteq \bigcup_{y \in Y} E_y,$$

which implies that p is an interior point of $\bigcup_{y \in Y} E_y$. Since p is arbitrary, $\bigcup_{y \in Y} E_y$ is open. $\square$

Corollary 2.4.12. *Let (X, d) be a metric space. Let $E_y \subseteq X$ be a closed set for each $y \in Y$. Then we have*

$$\bigcap_{y \in Y} E_y \text{ is closed}$$

whether Y is finite, countable, or uncountable. This corollary states that an arbitrary intersection of closed sets is closed.

Proof. By [Corollary 2.4.10](#), $(E_y)^c$ is open for each $y \in Y$. Then, by [Theorem 2.4.11](#), we have

$$\bigcup_{y \in Y} (E_y)^c \text{ is open.}$$

By [Theorem 2.4.9](#), we have

$$\bigcap_{y \in Y} E_y = \left(\bigcup_{y \in Y} (E_y)^c \right)^c \text{ is closed.}$$

$\square$

Theorem 2.4.13. *Let (X, d) be a metric space. Let $E_1, E_2, \dots, E_n \subseteq X$ be open sets for $n \in \mathbb{N}$. Then we have*

$$\bigcap_{k=1}^n E_k \text{ is open.}$$

This theorem states that a finite intersection of open sets is open.

Proof. Let $p \in \bigcap_{k=1}^n E_k$ be given. Then we have $p \in E_j$ for all $j = 1, 2, \dots, n$.

Since E_j is open, there exists some $r_j \in \mathbb{R}^+$ such that $N_{r_j}(p) \subseteq E_j$ for each $j = 1, 2, \dots, n$. Let R be a set defined as

$$R := \{r_j \mid j = 1, 2, \dots, n\} \subseteq \mathbb{R}^+.$$

Let $f : \mathbb{N}^n \rightarrow R$ be a function defined as

$$f(j) = r_j.$$

Then f is surjective. By [Lemma 2.2.1](#) and [theorem 2.2.4](#), R is finite. Then, by [Theorem 2.2.5](#), R has a minimum element, say $r = \min R > 0$. Then we have

$$N_r(p) \subseteq N_{r_j}(p) \subseteq E_j \text{ for all } j = 1, 2, \dots, n \implies N_r(p) \subseteq \bigcap_{k=1}^n E_k,$$

which implies that p is an interior point of $\bigcap_{k=1}^n E_k$. Since p is arbitrary, $\bigcap_{k=1}^n E_k$ is open. $\square$

Corollary 2.4.14. *Let (X, d) be a metric space. Let $E_1, E_2, \dots, E_n \subseteq X$ be closed sets for $n \in \mathbb{N}$. Then we have*

$$\bigcup_{k=1}^n E_k \text{ is closed.}$$

This corollary states that a finite union of closed sets is closed.

Proof. By [Corollary 2.4.10](#), $(E_k)^c$ is open for each $k = 1, 2, \dots, n$. Then, by [Theorem 2.4.13](#), we have

$$\bigcap_{k=1}^n (E_k)^c \text{ is open.}$$

By [Theorem 2.4.9](#), we have

$$\bigcup_{k=1}^n E_k = \left(\bigcap_{k=1}^n (E_k)^c \right)^c \text{ is closed.}$$

$\square$

Definition 2.4.15. Let (X, d) be a metric space and $E \subseteq X$. Let E' be the set of all limit points of E . The **closure** of E , which is denoted by $\overline{E}$, is defined as

$$\overline{E} := E \cup E'.$$

Theorem 2.4.16. *Let (X, d) be a metric space and $E \subseteq X$. Then we have the following properties:*

(A) $\overline{E}$ is closed.

(B) $E = \overline{E} \iff E$ is closed.

(C) For every closed subset $F \subseteq X$ such that $E \subseteq F$, we have $\overline{E} \subseteq F$.

(C) states that the closure of E is the smallest closed set containing E .

Proof. (A) We will show that $(\overline{E})^c$ is open. Let $p \in (\overline{E})^c$ be given. Since $p \notin E$ and $p \notin E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap \tilde{N}_r(p) = \emptyset \implies E \cap N_r(p) = \emptyset \implies N_r(p) \subseteq E^c.$$

By Theorem 2.4.7, $N_r(p)$ is open. Then, for every $q \in N_r(p)$, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$\begin{aligned} N_\varepsilon(q) \subseteq N_r(p) \subseteq E^c &\implies E \cap N_\varepsilon(q) = \emptyset \\ &\implies E \cap (\tilde{N}_\varepsilon(q) \cup \{q\}) = \emptyset \\ &\implies (E \cap \tilde{N}_\varepsilon(q)) \cup (E \cap \{q\}) = \emptyset \\ &\implies E \cap \tilde{N}_\varepsilon(q) = \emptyset \\ &\implies q \notin E'. \end{aligned}$$

Since q is arbitrary, we have

$$N_r(p) \subseteq (E')^c \implies N_r(p) \subseteq (\overline{E})^c = E^c \cap (E')^c,$$

which implies that p is an interior point of $(\overline{E})^c$. Since p is arbitrary, $(\overline{E})^c$ is open, which implies that $\overline{E}$ is closed by Corollary 2.4.10.

(B) E is closed $\iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$.

(C) Let $F \subseteq X$ be a closed set such that $E \subseteq F$. Let $p \in E'$ be given. Then, for every $r \in \mathbb{R}^+$, we have

$$E \cap \tilde{N}_r(p) \neq \emptyset \implies F \cap \tilde{N}_r(p) \neq \emptyset,$$

which implies that $p \in F' \subseteq F$. Therefore, we have $E' \subseteq F$. Since $E \subseteq F$, we have $\overline{E} = E \cup E' \subseteq F$. □

Theorem 2.4.17. Let $(\mathbb{R}, \|\cdot\|)$ be given. Let $E \subseteq \mathbb{R}$ be nonempty and bounded above. Then we have

$$\sup E \in \overline{E},$$

which implies that

$$E \text{ is closed} \implies \sup E \in E.$$

Proof. By the *least upper bound property*, there exists $\alpha = \sup E \in \mathbb{R}$. Assume that $\alpha \in E$. Then we have

$$E \subseteq \overline{E} \implies \alpha \in \overline{E}.$$

Otherwise, for every $x \in \mathbb{R}^+$, $\alpha - x$ is not an upper bound of E . Thus there exists some $p \in E$ such that

$$\alpha - x < p < \alpha \implies p \in \tilde{N}_x(\alpha) \cap E,$$

which implies that $\alpha \in E'$. Therefore, we have

$$E' \subseteq \overline{E} \implies \alpha \in \overline{E}.$$

□

Theorem 2.4.18. Let (X, d) be a metric space and $E \subseteq X$. Then we have

$$E \setminus E' = \{p \in X \mid p \text{ is an isolated point of } E\}.$$

Proof. ($\subseteq$) Let $p \in E \setminus E'$ be given. Since $p \notin E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap \tilde{N}_r(p) = \emptyset \implies (E \cap N_r(p)) \setminus \{p\} = \emptyset.$$

Since $p \in E$ and $p \in N_r(p)$, we have

$$E \cap N_r(p) = \{p\},$$

which implies that p is an isolated point of E .

($\supseteq$) Let p be an isolated point of E . Then there exists some $r \in \mathbb{R}^+$ such that

$$E \cap N_r(p) = \{p\}.$$

Then we have

- $E \cap \tilde{N}_r(p) = \emptyset \implies p \notin E'$.
- $p \in E$.

Therefore, we have $p \in E \setminus E'$.

□

Definition 2.4.19. Let (X, d) be a metric space and $E \subseteq X$. The **boundary** of E , which is denoted by ∂E , is defined as

$$\partial E := \overline{E} \cap \overline{E^c}.$$

Lemma 2.4.20. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for $n \in \mathbb{N}$. Then we have*

$$\tilde{N}_r(\mathbf{p}) \neq \emptyset \quad \text{for all } \mathbf{p} \in \mathbb{R}^n \text{ and } r \in \mathbb{R}^+.$$

Proof. Let $\mathbf{p} = (p_1, p_2, \dots, p_n) \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ be given. Then we have

$$d\left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1, \mathbf{p}\right) = \left\| \left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1\right) - \mathbf{p} \right\| = \left\| \frac{r}{2}\mathbf{e}_1 \right\| = \frac{r}{2}\|\mathbf{e}_1\| = \frac{r}{2} < r.$$

where $\mathbf{e}_1 = (1, 0, 0, \dots, 0) \in \mathbb{R}^n$. Therefore, we have

$$\mathbf{p} \neq \left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1\right) \in \tilde{N}_r(\mathbf{p}) \implies \tilde{N}_r(\mathbf{p}) \neq \emptyset.$$

□

Theorem 2.4.21. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have*

$$\partial E = (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)'),$$

which states that the boundary of E consists the following three disjoint sets:

- *The set of all isolated points of E .*
- *The set of all isolated points of E^c .*
- *The set of all limit points of both E and E^c .*

Proof. By the definition of the boundary, we have

$$\begin{aligned} \partial E &= \overline{E} \cap \overline{E^c} \\ &= (E \cup E') \cap (E^c \cup (E^c)') \\ &= (E \cap E^c) \cup (E \cap (E^c)') \cup (E' \cap E^c) \cup (E' \cap (E^c)'). \end{aligned}$$

Consider the following equivalences:

- $E = (E \setminus E') \cup (E \cap E')$.
- $E^c = (E^c \setminus (E^c)') \cup (E^c \cap (E^c)').$

Then we have the following equalities:

- $E \cap (E^c)' = ((E \setminus E') \cap (E^c)') \cup ((E \cap E') \cap (E^c)').$
- $E' \cap E^c = (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c \cap (E^c)')).$

Consider the last two terms of the above equalities:

- $((E \cap E') \cap (E^c)') \subseteq E' \cap (E^c)'.$

$$\bullet (E' \cap (E^c \cap (E^c)')) \subseteq E' \cap (E^c)'.$$

Then we have

$$\begin{aligned} \partial E &= \emptyset \cup ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)') \\ &= ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)') \end{aligned}$$

For every $\mathbf{p} \in E \setminus E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap \tilde{N}_r(\mathbf{p}) = \emptyset \implies \tilde{N}_r(\mathbf{p}) \subseteq E^c.$$

By Lemma 2.4.20, we have

$$\tilde{N}_s(\mathbf{p}) \cap \tilde{N}_r(\mathbf{p}) = \tilde{N}_{\min(s,r)}(\mathbf{p}) \neq \emptyset \quad \text{for all } s \in \mathbb{R}^+.$$

Then, for every $s \in \mathbb{R}^+$, we have

$$\begin{aligned} \tilde{N}_s(\mathbf{p}) \cap \tilde{N}_r(\mathbf{p}) \neq \emptyset &\implies \tilde{N}_s(\mathbf{p}) \cap (\tilde{N}_r(\mathbf{p}) \cap E^c) \neq \emptyset \\ &\implies \emptyset \neq (\tilde{N}_s(\mathbf{p}) \cap E^c) \cap \tilde{N}_r(\mathbf{p}) \subseteq \tilde{N}_s(\mathbf{p}) \cap E^c \\ &\implies E^c \cap \tilde{N}_s(\mathbf{p}) \neq \emptyset, \end{aligned}$$

which implies that $\mathbf{p} \in (E^c)'$. Then we have

$$E \setminus E' \subseteq (E^c)' \implies (E \setminus E') \cap (E^c)' = E \setminus E'.$$

Similarly, we can show that

$$E^c \setminus (E^c)' \subseteq E' \implies E' \cap (E^c \setminus (E^c)') = E^c \setminus (E^c)'.$$

Therefore, we have

$$\partial E = (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)').$$

□

Remark. Let (X, d) be a metric space and $Y \subseteq X$. Then, $(Y, d|_{Y \times Y})$ is also a metric space, since

$$d|_{Y \times Y} : Y \times Y \rightarrow \mathbb{R}$$

satisfies the properties of a metric. We call $(Y, d|_{Y \times Y})$ a **subspace** of (X, d) . For the sake of simplicity, we will denote $d|_{Y \times Y}$ by d and write (Y, d) instead of $(Y, d|_{Y \times Y})$.

Definition 2.4.22. Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. E is called **open relative** to Y if E is open in the subspace (Y, d) . That is, for every $p \in E$, there exists some $r \in \mathbb{R}^+$ such that

$$Y \cap N_r(p) \subseteq E.$$

Theorem 2.4.23. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. Then E is open relative to Y if and only if there exists an open set V in X such that $E = V \cap Y$.*

Proof. ($\Rightarrow$) Let E be an open set relative to Y . Then, for each $p \in E$, there exists some $r_p \in \mathbb{R}^+$ such that

$$Y \cap N_{r_p}(p) \subseteq E.$$

Let V be a set defined as

$$V := \bigcup_{p \in E} N_{r_p}(p).$$

By Theorems 2.4.7 and 2.4.11, V is open in X . Since $p \in N_{r_p}(p) \subseteq V$ for each $p \in E$, we have

- $E \subseteq Y$ and $E \subseteq V \Rightarrow E \subseteq V \cap Y$.
- $\bigcup_{p \in E} (Y \cap N_{r_p}(p)) \subseteq E \Rightarrow Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E \Rightarrow Y \cap V \subseteq E$.

Therefore, we have $E = V \cap Y$.

($\Leftarrow$) Let V be an open set in X such that $E = V \cap Y$. Then, for each $p \in E \subseteq V$, there exists some $r_p \in \mathbb{R}^+$ such that

$$N_{r_p}(p) \subseteq V \Rightarrow Y \cap N_{r_p}(p) \subseteq V \cap Y = E.$$

□

2.5 Compact Sets

Let (X, d) be a metric space and $E \subseteq X$. Let A be a set and let G_α be an open set in X for each $\alpha \in A$. Then the collection $\{G_\alpha\}_{\alpha \in A}$ is called an **open cover** of E if

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

A collection $\{G_\alpha\}_{\alpha \in S}$ is called a **subcover** of E if

- $S \subseteq A$.
- $E \subseteq \bigcup_{\alpha \in S} G_\alpha$.

E is called **compact** if every open cover of E has a finite subcover. That is, for each open cover $\{G_\alpha\}_{\alpha \in A}$ of E , there exists a subset $S_A \subseteq A$ such that

- S_A is finite.
- $E \subseteq \bigcup_{\alpha \in S_A} G_\alpha$.

Theorem 2.5.1. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. Then E is compact relative to Y if and only if E is compact relative to X .*

Proof. ($\Rightarrow$) Let E be a compact set relative to Y . Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of E relative to X . Then we have

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Let $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. By Theorem 2.4.23, H_α is open relative to Y . Since $E \subseteq Y$, we have

$$E \subseteq \left(\bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in A}$ is an open cover of E relative to Y . Since E is compact relative to Y , there exists a finite subcover $\{H_\alpha\}_{\alpha \in S_A}$ of E relative to Y . Then we have

$$E \subseteq \bigcup_{\alpha \in S_A} H_\alpha = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S_A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to X . Therefore, E is compact relative to X .

($\Leftarrow$) Let E be a compact set relative to X . Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of E relative to Y . Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha.$$

By Theorem 2.4.23, there exists an open set G_α in X such that $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha = \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in A}$ is an open cover of E relative to X . Since E is compact relative to X , there exists a finite subcover $\{G_\alpha\}_{\alpha \in S_A}$ of E relative to X . Then we have

$$E \subseteq \left(\bigcup_{\alpha \in S_A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) = \bigcup_{\alpha \in S_A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to Y . Therefore, E is compact relative to Y . $\square$

Remark. The concept of **compactness** is preserved under the subspace topology, while the concepts of **openness** and **closedness** are not, which is the reason why

“Let (X, d) be a metric space and $E \subseteq X$.”

this kind of statement is always included in theorems and definitions about metric spaces. In this textbook, you can assume that

- E is open. $\implies E$ is open relative to X .
- E is closed. $\implies E$ is closed relative to X .
- E is compact. $\implies E$ is compact relative to X .

if there is no specific mention of the subspace topology.

Theorem 2.5.2. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is closed.*

Proof. Let E be compact. We will prove that E^c is open, which implies that E is closed by [Corollary 2.4.10](#). Let $p \in E^c$ be given. Since $p \notin E$, we have

$$r_q := \frac{d(p, q)}{2} > 0 \quad \text{for each } q \in E.$$

Since $q \in N_{r_q}(q)$ for each $q \in E$, we have

$$E \subseteq \bigcup_{q \in E} N_{r_q}(q) \implies \{N_{r_q}(q)\}_{q \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{N_{r_q}(q)\}_{q \in S}$ of E . Let T be a set defined as

$$T := \{r_q \mid q \in S\}.$$

Let $f : S \rightarrow T$ be a function defined as

$$f(q) = r_q.$$

Then f is surjective. By [Theorem 2.2.4](#), T is finite. Then, by [Theorem 2.2.5](#), T has a minimum element, say $\varepsilon = \min T > 0$. For every $x \in N_\varepsilon(p)$, we have

$$d(x, p) < \varepsilon \leq r_q \quad \text{for each } q \in S.$$

Assume that there exists some $q \in S$ such that $x \in N_{r_q}(q)$. Then we have

$$d(p, q) \leq d(p, x) + d(x, q) < r_q + r_q = 2r_q = d(p, q),$$

which is a contradiction. Then the following statements hold:

- $N_\varepsilon(p) \cap N_{r_q}(q) = \emptyset$ for each $q \in S$.
- $E \subseteq \bigcup_{q \in S} N_{r_q}(q)$.

Therefore, we have $N_\varepsilon(p) \subseteq E^c$. Since p is arbitrary, E^c is open. $\square$

Theorem 2.5.3. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is bounded.*

Proof. Let E be compact. Let $p \in E$ be given. Let $\{G_n\}_{n \in \mathbb{N}}$ be a collection of open sets defined as

$$G_n = N_n(p) \quad \text{for each } n \in \mathbb{N}.$$

Then we have

$$n \leq m \implies G_n \subseteq G_m \quad \text{for each } n, m \in \mathbb{N}.$$

For every $q \in E$, by [Theorem 1.6.1](#), there exists $n \in \mathbb{N}$ such that $n > d(p, q)$. Thus we have

$$E \subseteq X \subseteq \bigcup_{n=1}^{\infty} G_n \implies \{G_n\}_{n \in \mathbb{N}} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_n\}_{n \in S}$ of E . Since $S \subseteq \mathbb{N}$ is finite, by [Theorem 2.2.5](#), S has a greatest element, say $m = \max S$. Then we have

$$E \subseteq \bigcup_{n \in S} G_n \subseteq G_m = N_m(p),$$

which implies that E is bounded. $\square$

Theorem 2.5.4. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then every closed subset $F \subseteq E$ is compact.*

Proof. Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of F . We have the following statements:

- F is closed. $\implies F^c$ is open.
- $F \subseteq \bigcup_{\alpha \in A} H_\alpha$.

Then we have

$$X = F \cup F^c \subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c = \bigcup_{\alpha \in A} (H_\alpha \cup F^c).$$

$$\bullet E \subseteq X \implies E \subseteq \bigcup_{\alpha \in A} (H_\alpha \cup F^c).$$

By Theorem 2.4.11, $H_\alpha \cup F^c$ is open for each $\alpha \in A$. Thus

$$\{H_\alpha \cup F^c\}_{\alpha \in A} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{H_\alpha \cup F^c\}_{\alpha \in B}$ of E . Then we have

$$\begin{aligned} \bullet E &\subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c). \\ \bullet F &\subseteq E \implies F \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c) = \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c. \end{aligned}$$

Since $F \cap F^c = \emptyset$, we have

$$F \subseteq \bigcup_{\alpha \in B} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in B}$ is a finite subcover of F . Therefore, F is compact. $\square$

Theorem 2.5.5. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then every infinite subset of E has a limit point in E .*

Proof. Let E be compact. Let $F \subseteq E$ be an infinite subset of E . Assume that F has no limit point in E . Then, for each $p \in E$, there exists $r_p \in \mathbb{R}^+$ such that

$$\tilde{N}_{r_p}(p) \cap F = \emptyset \implies N_{r_p}(p) \cap F \subseteq \{p\}.$$

Let $\{G_\alpha\}_{\alpha \in E}$ be a collection of open sets defined as

$$G_\alpha = N_{r_\alpha}(\alpha) \quad \text{for each } \alpha \in E.$$

Since $\alpha \in G_\alpha$ for each $\alpha \in E$, we have

$$E \subseteq \bigcup_{\alpha \in E} G_\alpha \implies \{G_\alpha\}_{\alpha \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_\alpha\}_{\alpha \in S}$ of E . Then we have

$$E \subseteq \bigcup_{\alpha \in S} G_\alpha \implies F \subseteq E \subseteq \bigcup_{\alpha \in S} G_\alpha.$$

Since $F \subseteq \bigcup_{\alpha \in S} G_\alpha$, we have

$$F = F \cap \bigcup_{\alpha \in S} G_\alpha = \bigcup_{\alpha \in S} (F \cap G_\alpha) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S,$$

which is a contradiction by Theorem 2.2.2. Therefore, every infinite subset of E has a limit point in E . $\square$

Theorem 2.5.6. *Let (X, d) be a metric space and let $\{K_\alpha\}_{\alpha \in A}$ be a collection of compact sets. Then we have*

$$\bigcap_{\alpha \in B} K_\alpha \neq \emptyset \text{ for all finite subset } B \subseteq A \implies \bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

Proof. Assume that $\bigcap_{\alpha \in A} K_\alpha = \emptyset$. Then we have

$$X = \left(\bigcap_{\alpha \in A} K_\alpha \right)^c = \bigcup_{\alpha \in A} (K_\alpha)^c.$$

Since each K_α is compact, K_α is closed by [Theorem 2.5.2](#). Then, by [Corollary 2.4.10](#), we have

$$\text{Each } (K_\alpha)^c \text{ is open} \implies \{(K_\alpha)^c\}_{\alpha \in A} \text{ is an open cover of } X.$$

Since $K_\alpha \subseteq X \subseteq \bigcup_{\beta \in A} (K_\beta)^c$, we have

$$\{(K_\beta)^c\}_{\beta \in A} \text{ is an open cover of } K_\alpha.$$

Since K_α is compact, there exists a finite subcover $\{(K_\beta)^c\}_{\beta \in B_\alpha}$ of K_α . Then we have

$$\begin{aligned} K_\alpha &\subseteq \bigcup_{\beta \in B_\alpha} (K_\beta)^c = \left(\bigcap_{\beta \in B_\alpha} K_\beta \right)^c \\ &\implies K_\alpha \cap \left(\bigcap_{\beta \in B_\alpha} K_\beta \right) = \emptyset \\ &\implies \bigcap_{\beta \in B_\alpha \cup \{\alpha\}} K_\beta = \emptyset. \end{aligned}$$

Since $\{\alpha\}$ and B_α are finite, $B_\alpha \cup \{\alpha\}$ is also finite by [Theorem 2.2.11](#), which contradicts the assumption. Therefore, we have

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

□

Lemma 2.5.7. *Let $(a_n)_{n=1}^\infty$ and $(b_n)_{n=1}^\infty$ be sequences in $\mathbb{R}$, and let $\{I_n\}_{n=1}^\infty$ be a sequence of intervals in $\mathbb{R}$ such that*

- $I_n = [a_n, b_n]$ for all $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$

where $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then we have

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Proof. Let E be a set defined as

$$E := \{a_n \mid n \in \mathbb{N}\}.$$

Since E is bounded above by b_1 , there exists $\alpha = \sup E$. Then we have

$$a_m \leq \alpha \quad \text{for all } m \in \mathbb{N}.$$

For every $m \in \mathbb{N}$, we have b_m is an upper bound of E since

- $a_n \leq a_m \leq b_m$ for all $n \in \mathbb{N}$ such that $n \leq m$.
- $a_n \leq b_n \leq b_m$ for all $n \in \mathbb{N}$ such that $n > m$.

Therefore, we have

$$a_m \leq \alpha \leq b_m \implies \alpha \in I_m \text{ for all } m \in \mathbb{N}.$$

□

Theorem 2.5.8. Let $k \in \mathbb{N}$ be given. Let $\{a_{n,j}\}_{n=1}^{\infty}$ and $\{b_{n,j}\}_{n=1}^{\infty}$ be sequences in $\mathbb{R}$ for all $j = 1, 2, \dots, k$, and let $\{I_n\}_{n=1}^{\infty}$ be a sequence of k -cells in $\mathbb{R}^k$ such that

- $I_n = \left\{ \mathbf{x} \in \mathbb{R}^k \mid a_{n,j} \leq x_j \leq b_{n,j} \text{ for all } j = 1, 2, \dots, k \right\}$ for all $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$

where $a_{n,j} \leq b_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$. Then we have

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Proof. Let $\{I_{n,j}\}_{n=1}^{\infty}$ be a sequence of intervals in $\mathbb{R}$ defined as

$$I_{n,j} = [a_{n,j}, b_{n,j}] \quad \text{for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}.$$

Then we have $I_{n+1,j} \subseteq I_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$. By Lemma 2.5.7, we have

$$\bigcap_{n=1}^{\infty} I_{n,j} \neq \emptyset \quad \text{for all } j = 1, 2, \dots, k.$$

Let $\alpha_j \in \bigcap_{n=1}^{\infty} I_{n,j}$ be given for all $j = 1, 2, \dots, k$. Then we have

$$\alpha_j \in I_{n,j} \implies a_{n,j} \leq \alpha_j \leq b_{n,j} \quad \text{for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}.$$

Let $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k) \in \mathbb{R}^k$. Then we have

$$\alpha \in I_n \quad \text{for all } n \in \mathbb{N} \implies \alpha \in \bigcap_{n=1}^{\infty} I_n.$$

□

Theorem 2.5.9. Let $(\mathbb{R}^k, \|\cdot\|)$ be given for $k \in \mathbb{N}$. Then every k -cell is compact.

Proof. Let I be a k -cell in $\mathbb{R}^k$, and let $I_1 := I$. Then we have

$$I_1 = \prod_{j=1}^k [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k,$$

where $-\infty < a_{1,j} \leq b_{1,j} < +\infty$ for all $j = 1, 2, \dots, k$. Let $I_{1,j,m}$ be an interval in $\mathbb{R}$ defined as

$$I_{1,j,m} := \begin{cases} \left[a_{1,j}, \frac{a_{1,j} + b_{1,j}}{2} \right] & \text{if } m = 1 \\ \left[\frac{a_{1,j} + b_{1,j}}{2}, b_{1,j} \right] & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \dots, k.$$

Let S_k be a set defined as

$$S_k := \{1, 2\}^k = \left\{ \mathbf{m} \in \mathbb{R}^k \mid m_j \in \{1, 2\} \text{ for all } j = 1, 2, \dots, k \right\}.$$

By Theorem 2.2.13, S_k is finite. We have

$$I_1 = \bigcup_{\mathbf{m} \in S_k} \left(\prod_{j=1}^k I_{1,j,m_j} \right).$$

Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of I_1 . Assume that I_1 has no finite subcover. Then, for each $\mathbf{m} \in S_k$, we have

$$\prod_{j=1}^k I_{1,j,m_j} \subseteq I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of $\prod_{j=1}^k I_{1,j,m_j}$. Assume that for each $\mathbf{m} \in S_k$, we have

$$\prod_{j=1}^k I_{1,j,m_j} \text{ has a finite subcover } \{G_\alpha\}_{\alpha \in A_{\mathbf{m}}}.$$

Then we have

$$\prod_{j=1}^k I_{1,j,m_j} \subseteq \bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha \quad \text{for all } \mathbf{m} \in S_k,$$

where $\bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha$ is open by [Theorem 2.4.11](#). Then we have

$$I_1 \subseteq \bigcup_{\mathbf{m} \in S_k} \left(\bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha \right) = \bigcup_{\alpha \in A} G_\alpha,$$

where $A := \bigcup_{\mathbf{m} \in S_k} A_{\mathbf{m}}$ is finite by [Theorem 2.2.11](#). Thus I_1 has a finite subcover, which is a contradiction. Therefore, there exists $\mathbf{m}_1 \in S_k$ such that

$$\prod_{j=1}^k I_{1,j,(m_1)_j} \text{ has no finite subcover.}$$

Let I_2 be a k -cell in $\mathbb{R}^k$ defined as

$$I_2 := \prod_{j=1}^k [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k,$$

where $a_{2,j}, b_{2,j} \in \mathbb{R}$ is defined as

$$\begin{aligned} \bullet \quad a_{2,j} &:= \begin{cases} a_{1,j} & \text{if } (m_1)_j = 1 \\ \frac{a_{1,j} + b_{1,j}}{2} & \text{if } (m_1)_j = 2 \end{cases} \\ \bullet \quad b_{2,j} &:= \begin{cases} \frac{a_{1,j} + b_{1,j}}{2} & \text{if } (m_1)_j = 1 \\ b_{1,j} & \text{if } (m_1)_j = 2 \end{cases} \end{aligned}$$

for all $j = 1, 2, \dots, k$. Then, for each $j = 1, 2, \dots, k$, we have

$$[a_{2,j}, b_{2,j}] = I_{1,j,(m_1)_j} \subseteq [a_{1,j}, b_{1,j}],$$

which implies that

$$I_2 = \prod_{j=1}^k I_{1,j,(m_1)_j} \subseteq I_1.$$

Let $I_{2,j,m}$ be an interval in $\mathbb{R}$ defined as

$$I_{2,j,m} := \begin{cases} \left[a_{2,j}, \frac{a_{2,j} + b_{2,j}}{2} \right] & \text{if } m = 1 \\ \left[\frac{a_{2,j} + b_{2,j}}{2}, b_{2,j} \right] & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \dots, k.$$

Then we have

$$I_2 = \bigcup_{\mathbf{m} \in S_k} \left(\prod_{j=1}^k I_{2,j,m_j} \right).$$

We have shown that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of I_2 , and I_2 has no finite subcover. Then, for each $\mathbf{m} \in S_k$, we have

$$\prod_{j=1}^k I_{2,j,m_j} \subseteq I_2 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of $\prod_{j=1}^k I_{2,j,m_j}$. Therefore, there exists $\mathbf{m}_2 \in S_k$ such that

$$\prod_{j=1}^k I_{2,j,(\mathbf{m}_2)_j} \text{ has no finite subcover.}$$

By repeating this process, we can construct the following sequences:

- The sequence $(\mathbf{m}_n)_{n=1}^\infty$ in S_k .
- The sequences $(a_{n,j})_{n=1}^\infty$ and $(b_{n,j})_{n=1}^\infty$ in $\mathbb{R}$ for all $j = 1, 2, \dots, k$.
- The sequence $(I_{n,j,m})_{n=1}^\infty$ of intervals in $\mathbb{R}$ for all $j = 1, 2, \dots, k$ and $m = 1, 2$.
- The sequence $(I_n)_{n=1}^\infty$ of k -cells in $\mathbb{R}^k$.

We have the following properties:

- $I_{n+1} \subseteq I_n$ for all $n \in \mathbb{N}$.
- I_n has no finite subcover for all $n \in \mathbb{N}$.
- $a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$.
- $b_{n+1,j} - a_{n+1,j} = \frac{1}{2}(b_{n,j} - a_{n,j}) \implies b_{n,j} - a_{n,j} = \frac{1}{2^{n-1}}(b_{1,j} - a_{1,j})$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$.

Let $\delta := \sqrt{\sum_{j=1}^k (b_{1,j} - a_{1,j})^2}$. Then we have

$$\|x - y\| \leq \sqrt{\sum_{j=1}^k (b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}} \quad \text{for all } x, y \in I_n \text{ and } n \in \mathbb{N}.$$

By [Theorem 2.5.8](#), we have $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$. Let $\beta \in \bigcap_{n=1}^{\infty} I_n$ be given. Then there exists $\alpha_0 \in A$ such that $\beta \in G_{\alpha_0}$. Since G_{α_0} is open, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$B_\varepsilon(\beta) \subseteq G_{\alpha_0}.$$

By [Corollary 1.6.2](#), there exists $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \frac{\varepsilon}{2\delta}$. Since $\beta \in I_N$, we have

$$\|\beta - \gamma\| \leq \frac{\delta}{2^{N-1}} < \varepsilon \quad \text{for all } \gamma \in I_N,$$

which implies that $I_N \subseteq B_\varepsilon(\beta) \subseteq G_{\alpha_0}$. Therefore, I_N has a finite subcover $\{G_{\alpha_0}\}$, which is a contradiction. Thus I has a finite subcover, and I is compact. $\square$

Corollary 2.5.10. *For every $-\infty < a \leq b < +\infty$, the closed interval $[a, b]$ is compact in $(\mathbb{R}, \|\cdot\|)$.*

Theorem 2.5.11. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have*

$$E \text{ is compact} \iff E \text{ is closed and bounded.}$$

*This theorem is called the **Heine-Borel Theorem**.*

Proof. ($\implies$) Let E be compact. E is closed by [Theorem 2.5.2](#), and E is bounded by [Theorem 2.5.3](#).

($\impliedby$) Let E be closed and bounded. Then there exists some $p \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ such that $E \subseteq B_r(p)$. Let I be a n -cell in $\mathbb{R}^n$ defined as

$$I := \prod_{j=1}^n [p_j - r, p_j + r] \subseteq \mathbb{R}^n.$$

Then we have $E \subseteq B_r(p) \subseteq I$. By [Theorem 2.5.9](#), I is compact. Since E is closed, by [Theorem 2.5.4](#), E is compact. $\square$

2.6 Connected Sets

Let (X, d) be a metric space and $E \subseteq X$. E is called **connected** if there exists no pair of sets $A, B \subseteq X$ such that

- $\emptyset \neq A \subseteq E$ and $\emptyset \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$.

Such pair of sets A and B is called **separated**. If E is not connected, then E is called **disconnected**.

Theorem 2.6.1. *Let $(\mathbb{R}, \|\cdot\|)$ be given. Then $E \subseteq \mathbb{R}$ is connected if and only if*

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Proof. ($\implies$) Let $E \subseteq \mathbb{R}$ be connected. Assume that there exist $a, b \in E$ and $x \in \mathbb{R}$ such that

$$a < x < b \text{ and } x \notin E.$$

Let A, B be sets defined as

- $A := E \cap (-\infty, x) \subseteq E$.
- $B := E \cap (x, \infty) \subseteq E$.

Then we have

- $A \neq \emptyset$ and $B \neq \emptyset$ since $a \in A$ and $b \in B$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$ since $x \notin E$.

Thus A and B are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

($\impliedby$) Let $E \subseteq \mathbb{R}$ be a set such that

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume that E is disconnected. Then there exist separated sets $A, B \subseteq \mathbb{R}$ such that

- $\emptyset \neq A \subseteq E$ and $\emptyset \neq B \subseteq E$.

- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$.

Let $a \in A$ and $b \in B$ be given. Without loss of generality, assume that $a < b$. Let

$$\alpha := \sup(A \cap [a, b]) \in \mathbb{R},$$

where we have

- $a \in A \implies A \cap [a, b] \neq \emptyset$.
- $A \cap [a, b]$ is bounded above by b .

Then the *existence* of α is guaranteed by the *least upper bound property*. By [Theorem 2.4.17](#), we have

$$\alpha \in \overline{(A \cap [a, b])} = (A \cap [a, b]) \cup (A \cap [a, b])'.$$

By [Definition 2.4.4](#), it is clear that

$$(A \cap [a, b])' \subseteq A'.$$

Thus we have

$$\alpha \in (A \cap [a, b]) \subseteq A \quad \text{or} \quad \alpha \in (A \cap [a, b])' \subseteq A' \implies \alpha \in \overline{A}.$$

Then the following statements hold:

- $\alpha = \sup(A \cap [a, b]) \implies a \leq \alpha \leq b$.
- $\alpha \in \overline{A} \implies \alpha \notin B \implies \alpha \neq b$.

If $\alpha \notin A$, then we have

- $\alpha \neq a \implies a < \alpha < b$
- $\alpha \notin A$ and $\alpha \notin B \implies \alpha \notin E$

which is a contradiction. Otherwise, we have

$$\alpha \in A \implies \alpha \notin \overline{B}.$$

Then there exists $r \in (0, b - \alpha)$ such that

$$\tilde{N}_r(\alpha) \cap B = \emptyset \implies \beta := \alpha + r \notin B.$$

Then we have

- $a \leq \alpha < \beta < b$
- $\beta > \alpha \implies \beta \notin A \implies \beta \notin E$

which is a contradiction. Therefore, we conclude that E is connected. $\square$

Chapter 3

Sequences and Series

3.1 Convergent Sequences

Let (X, d) be a metric space. A **sequence** $(x_n)_{n=1}^{\infty}$ is a function $f : \mathbb{N} \rightarrow X$, where $x_n = f(n)$ for each $n \in \mathbb{N}$.

A sequence $(x_n)_{n=1}^{\infty}$ is said to be **convergent** if there exists $x \in X$ such that

For every $\varepsilon \in \mathbb{R}^+$, there exists $N \in \mathbb{N}$ such that $n \geq N \implies d(x_n, x) < \varepsilon$.

In this case, we say that

- $(x_n)_{n=1}^{\infty}$ **converges** to x .
- x is the **limit** of $(x_n)_{n=1}^{\infty}$.

We denote the limit of $(x_n)_{n=1}^{\infty}$ by

- $\lim_{n \rightarrow \infty} x_n = x$.
- $x_n \rightarrow x$ as $n \rightarrow \infty$.

If $(x_n)_{n=1}^{\infty}$ is not convergent, it is said to be **divergent**.

Example 3.1.1. Let $(\mathbb{R}, \|\cdot\|)$ be given. Let $(x_n)_{n=1}^{\infty}$ be a sequence defined as

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

Then, for each $\varepsilon \in \mathbb{R}^+$, we have

$$n > \frac{1}{\varepsilon} \implies \|x_n - 0\| = \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \varepsilon.$$

Therefore, $(x_n)_{n=1}^{\infty}$ converges to 0.

Let $(\mathbb{R}^+, \|\cdot\|)$ be a subspace of $(\mathbb{R}, \|\cdot\|)$. Let $(x_n)_{n=1}^\infty$ be a sequence defined as

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

By [Theorem 1.6.1](#), for each $y \in \mathbb{R}^+$, there exists $N_y \in \mathbb{N}$ such that $N_y > \frac{1}{y}$. Let

$$\varepsilon_y := y - \frac{1}{N_y} \in \mathbb{R}^+ \quad \text{for each } y \in \mathbb{R}^+.$$

Then we have

$$n > N_y \implies \|x_n - y\| = \left| \frac{1}{n} - y \right| > \left| \frac{1}{N_y} - y \right| = \varepsilon_y.$$

Therefore, $(x_n)_{n=1}^\infty$ diverges in $(\mathbb{R}^+, \|\cdot\|)$.

Remark. Let (X, d) be a metric space. In this textbook, $(x_n)_{n=1}^\infty$ represents a sequence in (X, d) , while $\{x_n\}_{n=1}^\infty$ represents a set defined as

$$\{x_n\}_{n=1}^\infty := \{x_n \mid n \in \mathbb{N}\} \subseteq X.$$

Definition 3.1.2. Let (X, d) be a metric space. A sequence $(x_n)_{n=1}^\infty$ is said to be **Bounded** if $\{x_n\}_{n=1}^\infty$ is bounded.

Theorem 3.1.3. Let (X, d) be a metric space. Let $(x_n)_{n=1}^\infty$ be a convergent sequence. Then we have

- (A) $(x_n)_{n=1}^\infty$ is bounded.
- (B) The limit of $(x_n)_{n=1}^\infty$ is unique.

Proof. Let x be the limit of $(x_n)_{n=1}^\infty$.

- (A) Let $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, x) < 1.$$

Let S be a set defined as

$$S := \{d(x_n, x) \mid 1 \leq n \leq N\} \subseteq [0, \infty).$$

Let $f : \mathbb{N}^N \rightarrow S$ be a function defined as

$$f(k) = d(x_k, x) \quad \text{for all } k \in \mathbb{N}^N.$$

Then f is surjective. By [Lemma 2.2.3](#), S is finite. Then, by [Theorem 2.2.5](#), S has a greatest element, say $M = \max S$. Then we have

$$d(x_n, x) < M + 1 \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $(x_n)_{n=1}^\infty$ is bounded.

(B) Let y be the limit of $(x_n)_{n=1}^\infty$. Assume that $x \neq y$. Then we have $r := d(x, y) > 0$. Let $\varepsilon = \frac{r}{2}$. Then we have

- There exists $N_x \in \mathbb{N}$ such that $n \geq N_x \implies d(x_n, x) < \frac{r}{2}$.
- There exists $N_y \in \mathbb{N}$ such that $n \geq N_y \implies d(x_n, y) < \frac{r}{2}$.

Thus we have

$$n \geq \max\{N_x, N_y\} \implies d(x, y) \leq d(x, x_n) + d(x_n, y) < r,$$

which is a contradiction. Therefore, we have $x = y$.

□

Theorem 3.1.4. *Let (X, d) be a metric space. Then the following statements hold:*

(A) *A sequence $(x_n)_{n=1}^\infty$ converges to $x \in X$ if and only if*

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^\infty$.

(B) *Let $E \subseteq X$ be given. If x is a limit point of E , then there exists a sequence $(x_n)_{n=1}^\infty$ in (X, d) such that*

$$\lim_{n \rightarrow \infty} x_n = x.$$

Proof. (A) ($\implies$) Let $(x_n)_{n=1}^\infty$ converge to $x \in X$. Then, for each $\varepsilon \in \mathbb{R}^+$, there exists $M_\varepsilon \in \mathbb{N}$ such that

$$n \geq M_\varepsilon \implies d(x_n, x) < \varepsilon,$$

which implies that

$$\{x_n\}_{n=M_\varepsilon}^\infty \subseteq N_\varepsilon(x).$$

($\impliedby$) Let $(x_n)_{n=1}^\infty$ be a sequence in (X, d) such that

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^\infty$.

(B) Let $E \subseteq X$ be given. Assume that x is a limit point of E . Then for every $\varepsilon \in \mathbb{R}^+$, we have $(N_\varepsilon(x) - \{x\}) \cap E \neq \emptyset$. Thus we can choose a sequence $(y_n)_{n=1}^\infty$ in (X, d) such that

$$y_n \in (N_{\frac{1}{n}}(x) \setminus \{x\}) \cap E.$$

Then we have

$$d(y_n, x) < \frac{1}{n} \quad \text{for each } n \in \mathbb{N},$$

which implies that $\lim_{n \rightarrow \infty} y_n = x$.

□

- 3.2 Subsequences**
- 3.3 Cauchy Sequences**
- 3.4 Upper and Lower Limits**
- 3.5 Some Special Sequences**
- 3.6 Series**
- 3.7 Series of Nonnegative Terms**
- 3.8 The Number e**
- 3.9 The Root and Ratio Tests**
- 3.10 Power Series**
- 3.11 Summation by Parts**
- 3.12 Absolute Convergence**
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7.7 Equicontinuous Families of Functions

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11.6 Integration

11.7 Comparison with the Riemann Integral

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11.9 Functions of Class $\mathcal{L}^2$

Appendix: Construction of $\mathbb{R}$

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